CivWorldBench: Evaluating Language World Models in Structured Multi-Agent Sandbox Worlds

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Abstract

The growing use of language models for forecasting motivates a broader evaluation of their world-model capabilities. Such evaluation must go beyond predictive accuracy to assess hidden-state inference and intervention prediction. Real-world forecasting benchmarks offer limited access to hidden-state ground truth and controlled experiments for evaluating intervention predictions. We introduce CivWorldBench, a benchmark built on a multi-agent strategy game engine with recorded internal states, controlled observations, and paired intervention rollouts. Its 1,900 decision points from 475 worlds support three task levels: association forecasting, belief inference, and intervention prediction. Controlled observations and reference predictors support attributable evaluation of capability gaps, with an expert harness and empirical ceiling estimates contextualizing scores. Stored trajectories make intermediate predictions verifiable, while agent-play experiments test whether world-model capabilities are usable for decision-making. The main panel comprises eleven frontier language models spanning different capability levels. Composite benchmark scores correlate with real-world forecasting leaderboards on matched model panels, more closely than with a general-capability index, supporting its use as a proxy for real-world forecasting capability. Strong association forecasts do not consistently extend to belief inference or intervention prediction, and effect magnitudes remain poorly calibrated despite informative direction predictions. Process verification reveals errors missed by final-answer evaluation, and structured reasoning improves intervention prediction for the strongest model in the process study. Agent-play experiments identify useful assisted plans and further motivate evaluating decision-relevant capabilities beyond association forecasting. These findings identify hidden-state inference, intermediate-state prediction, and effect calibration as priorities for world-model training.

Homepage: https://leon-francis.github.io/CivWorldBench/

Code: https://github.com/Leon-Francis/CivWorldBench

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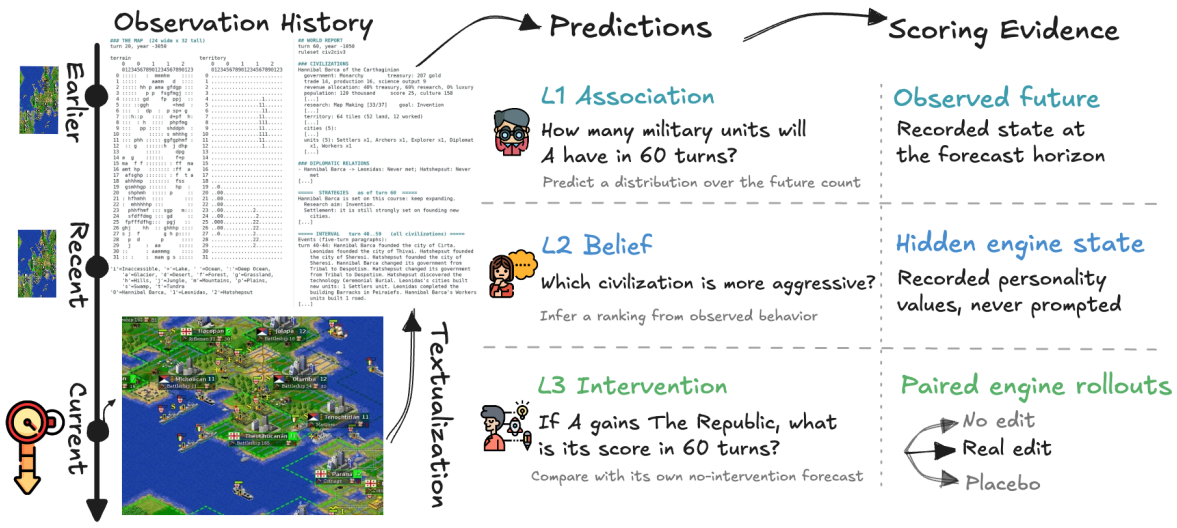


Figure 1: Overview of CIVWORLDBENCH. Textual observation histories support association forecasting, belief inference, and intervention prediction. Predictions are evaluated against recorded futures, hidden engine states, and paired engine rollouts, respectively.

1 Introduction

Language models increasingly support forecasting and decision-making in multi-agent systems. They simulate agent populations (Park et al., 2023; Piatti et al., 2024; Piao et al., 2025), forecast geopolitical and economic events (Halawi et al., 2024; Karger et al., 2025; Xu et al., 2026), negotiate (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022; Zhou et al., 2024), and act in digital, social, and physical environments (Zhu & Cai, 2026). These applications require a *world model* that predicts observable dynamics, infers relevant hidden state, and anticipates the consequences of interventions (Chen et al., 2026; Chu et al., 2026). Forecasting is the observable output of such a world model: a forecast commits implicitly to beliefs about hidden state and dynamics, but its accuracy alone does not establish that these beliefs are correct (Vafa et al., 2024), and intervention prediction further requires mechanism assumptions beyond observed associations (Pearl, 2009; Bareinboim et al., 2022).

Existing benchmarks each isolate part of this problem. Real-world forecasting benchmarks such as ForecastBench (Karger et al., 2025) resolve against genuine outcomes but provide no latent-state ground truth and no alternative trajectories, so belief and dynamics errors cannot be separated. Causal-reasoning benchmarks test inference from statistical descriptions (Jin et al., 2024) or interactive discovery in synthetic laboratories (Yang et al., 2026a), and SocietyBench forecasts social events on anonymized timelines (Wang et al., 2026b). None of them pairs forecasting, hidden-state inference, and intervention prediction on the same recorded multi-agent dynamics. Game-agent benchmarks such as CivRealm and BALROG (Qi et al., 2024; Paglieri et al., 2025) score an acting policy, entangling world knowledge with planning and execution. Isolating world-model capabilities therefore requires prediction tasks with verifiable hidden states and intervention outcomes, plus controls for observation limits and stochasticity.

We introduce CIVWORLDBENCH, a benchmark built on the Freeciv game engine.[1] Civilizations expand, research, trade, cooperate, and compete under recorded rules and latent behavioral parameters. Figure 1 illustrates how textual observation histories support three task levels: L1 association forecasting (QUANTITY, EVENT), L2 belief inference (TRAIT, RULES), and L3 intervention prediction (INTERVENE). The benchmark contains 475 worlds, 1,900 decision points, and 18,499 prompts. Global, egocentric, and out-of-distribution settings separate observation limits from configuration shifts. Saved checkpoints en-

[1]Freeciv is an open-source turn-based strategy game, `https://www.freeciv.org`.

able paired intervention, control, and placebo rollouts, and re-seeded futures measure engine stochasticity. FLOOR, EXPERT, and CEILING references contextualize performance, and the Sandbox-World Forecasting Index (SWFI) aggregates reference-normalized scores across levels and settings.

Our evaluation of eleven language models identifies distinct capability profiles. SWFI correlates with real-world forecasting leaderboards on matched model panels more closely than with a general-capability index, supporting its use as a forecasting-specific proxy. Within the benchmark, association forecasting rankings do not consistently reflect belief and intervention capabilities. Most models predict intervention directions above chance while estimating effect magnitudes poorly. Process verification identifies intermediate errors missed by final-answer scores, and the strongest model in the process study improves effect prediction through structured reasoning. Agent-play results show that useful advice is not identified by association forecasting alone.

Our contributions are threefold. First, we enable *attributable evaluation* through a structured multi-agent benchmark with controlled observations, paired intervention outcomes, and task-specific reference predictors. Second, we make answers and intermediate predictions *verifiable* against recorded engine states and rollouts, including Table and Code reasoning protocols. Third, we assess whether these capabilities are *usable* in agent decision-making and how they relate to real-world forecasting. Engine patches and evaluation code are available at `https://github.com/Leon-Francis/CivWorldBench`. The benchmark data release is in preparation.

2 The CivWorldBench Environment

CIVWORLDBENCH uses Freeciv to combine multi-agent dynamics with recorded hidden states, controlled observations, and executable interventions. Each world contains 3–10 AI civilizations that expand, research, trade, form alliances, and fight over hundreds of turns.

2.1 State, dynamics, and observations

A world $w \sim \mathcal{W}$ is an engine instance with state $s_t = (x_t, z_t; \theta_w)$, where x_t contains reportable map, city, unit, economic, technological, and diplomatic fields, the hidden dynamic state z_t includes attitudes and internals, and θ_w contains fixed personality traits and world parameters. Formally, each world is a partially observable Markov decision process $(\mathcal{S}, \mathcal{A}, T, \Omega, O_v)$ without a reward, since the benchmark evaluates prediction rather than control. Each built-in agent i follows a policy π_i that maps its observation $O_{\text{ego}(i)}(s_t) \in \Omega$ to an action in $\mathcal{A}$, and the joint actions together with the random innovation u_t induce the transition $s_{t+1} = T(s_t, \{\pi_i\}, u_t)$. The evaluated world model is a map $f(o_{\le t_1}, \{\pi_i\}) \mapsto (b(z_{t_1}, \theta_w),\ \hat{p}(s_{t_1+h}))$ from the observation history and the agents' current policies to a belief over the hidden state and a forecast of the future state under those policies.

The world model's input at t_1 is three-frame history $o_{\le t_1} = (o_{t_1-2\Delta}, o_{t_1-\Delta}, o_{t_1}, \text{narrative}, \text{policies})$ with $\Delta = 40$ turns, or 20 when $t_1 < 81$, where each frame $o_t = O_v(s_t)$ is a text report. The global view $v = \text{omni}$ renders reportable fields across civilizations, and the egocentric view $v = \text{ego}(i)$ is the engine's own fog-of-war rendering for agent i. The narrative summarizes observed events in five-turn blocks, and the policies give each agent's research aim and course of action, such as its expansion or war posture. The output is elicited as quantile forecasts, event probabilities, and hidden-state beliefs, as Section 3 specifies. Section C gives the formal definitions.

2.2 Behavioral validity of hidden targets

Hidden state includes personality traits that shape strategic preferences, diplomatic attitudes that influence relations between civilizations, and world parameters governing disaster frequency, city growth, and research speed. L2 infers traits and world parameters, and L3 additionally intervenes on attitudes. In Freeciv's default AI, several traits had little or no effect and war was rarely reversed. We apply 5

Table 2: Task definitions and evaluation metrics. Prompt counts include all settings and ablation conditions. Reference ceilings are for OMNI-ID.

Level	Task	Prediction target	Ground truth	Metric (reference → ceiling)	Prompts
L1	QUANTITY	Five quantiles for 11 quantities at two horizons	Mainline	CRPS skill over trend ($0 \rightarrow +0.68$)	3,600
	EVENT	Probabilities for 11 event templates: ten binary with supplied base rates, one leader	Mainline	Brier gain over reference ($0 \rightarrow +0.11$)	3,600
L2	TRAIT	Rank civilizations on four traits	Recorded θ_w	Adjusted pairwise accuracy ($0 \rightarrow 0.66$)	3,360
	RULES	Infer three hidden world rules from observable evidence	Recorded θ_w	Adjusted accuracy ($0 \rightarrow 0.93$)	
L3	INTERVENE	Quantity forecasts for 11 edits and matched placebos	Paired rollouts	Capture, adjusted direction ($0 \rightarrow 0.92$), specificity	7,939

source patches to activate these traits, initialize contact at peace, and permit ceasefires. Sections D and K document the modifications.

2.3 World generation

Table 1: Corpus composition across 475 worlds and 1,900 decision points, with 561,301 per-turn state records and 7,104 collected arms.

Set	Class	Worlds	Points	Agents	Map	Turns
Basic	small	75	300	3	24×32	400
	medium	75	300	6	32×64	800
	large	75	300	8	48×80	1200
Gen.	joint	250	1,000	3–10	24×32–64×85	400–1200

Table 1 summarizes the two world sets. The *basic* set fixes the ruleset and difficulty across three map classes while varying hidden parameters and generated worlds. The *generalization* set jointly varies the ruleset, AI difficulty, map size, civilization count, hidden-parameter ranges, and team play: 128 worlds use the classic ruleset, which changes the unit, building, and technology sets, five difficulty levels appear, and 60 worlds play in teams, so the shift affects game dynamics rather than only the map. Each world contributes four decision points, and collection runs beyond the reporting budget to record forecast targets. Median QUANTITY prompt lengths are 17.9K, 47.6K, and 78.1K tokens for the basic map classes.

2.4 Interventions

An intervention applies an edit m to a saved checkpoint before continuing the simulation. Each eligible checkpoint produces four arms: an unedited *control*, a repeated *recheck*, a real edit, and a matched *placebo* m_0 with negligible or mechanism-irrelevant impact. Real-arm outcomes are compared with the rerun control rather than the mainline, and placebos measure responses to nominal edits. Reloads and edits can alter random-number consumption, so branch differences retain stochastic variation, as Section D.3 documents.

3 Tasks and Evaluation Settings

The three levels query different components of the process in Section 2: L1 predicts future observables x_{t+h}, L2 infers the static hidden parameters θ_w, and L3 predicts x_{t+h} after an external edit to s_t. Table 2 summarizes the five tasks.

L1: Association. QUANTITY requests five quantiles for eleven quantities at two budget-dependent horizons, listed in Section E. Six quantity families, namely economy, scale, technology–culture, military, diplomacy, and overall score, receive equal weight. EVENT requests probabilities for ten binary event templates, including city capture, wonder completion, and alliance formation, and a distribution over the future score leader.

L2: Belief. TRAIT ranks civilizations on aggression, expansion, trade, and building preference. RULES classifies disaster frequency, growth speed, and research speed into three bins each, subject to evidence requirements. Both tasks use recorded engine values as ground truth.

L3: Intervention. Eleven operators modify treasury, research points, cities, government access, map visibility, personality traits, attitudes, or disaster frequency. The model receives the history and edit description, then re-answers QUANTITY within the operator's target and horizon range. *Capture* compares the predicted shift from the model's own control answer with the realized shift between engine branches. *Direction* assesses sign agreement, and *specificity* measures response selectivity between real and placebo edits.

Settings. OMNI-ID uses the basic worlds with global observations. EGO uses the same worlds through one civilization's view, with questions scoped to contacted civilizations. OMNI-OOD applies global observations to the generalization worlds, whose rulesets, AI difficulty, map sizes, civilization counts, hidden-parameter ranges, and team structures differ from the basic set, separating restricted information from shifts in world dynamics. Table 17 reports information ablations.

4 Evaluation Protocol and Research Questions

4.1 Scores

Quantity forecasting. For each cell c, indexed by world, decision point, civilization, quantity, and horizon, let y be the engine target and $\hat{q} = (\hat{q}_\tau)_{\tau\in\mathcal{T}}$ the predicted quantiles at $\mathcal{T} = \{0.1, 0.25, 0.5, 0.75, 0.9\}$. Quantile loss (QL) approximates the continuous ranked probability score (CRPS) (Gneiting & Raftery, 2007):

$$\mathrm{QL}(\hat{q}, y) = \frac{2}{|\mathcal{T}|} \sum_{\tau\in\mathcal{T}} \rho_\tau\big(\varphi(y) - \varphi(\hat{q}_\tau)\big), \qquad \rho_\tau(u) = \big(\tau - \mathbb{1}[u < 0]\big)u, \tag{1}$$

where ρ_τ is the pinball loss and $\mathbb{1}[\cdot]$ the indicator. The transform $\varphi(x) = \mathrm{sign}(x)\log(1 + |x|)$ applies to treasury and population levels and culture increments, and is the identity otherwise. The normalized cell loss is $\ell_c = \mathrm{QL}(\hat{q}, y)/\sigma_v$, where $\sigma_v > 0$ is the fixed corpus scale for quantity v, equal to one for log-transformed quantities. Enemy and ally counts instead use the absolute error of the predicted median.

For eligible cells $\mathcal{C}_{wf}$ of family f in world w, let $\ell_{c,\mathrm{ref}}$ denote trend-reference loss. World–family skill S_{wf} and the Quantity score S_{QNT} are

$$S_{wf} = \mathrm{clip}_{[-1,1]}\left(1 - \frac{\sum_{c\in\mathcal{C}_{wf}} \ell_c}{\sum_{c\in\mathcal{C}_{wf}} \ell_{c,\mathrm{ref}}}\right), \qquad S_{\mathrm{QNT}} = \frac{1}{|\mathcal{W}|} \sum_{w\in\mathcal{W}} \frac{1}{|\mathcal{F}_w|} \sum_{f\in\mathcal{F}_w} S_{wf}, \tag{2}$$

where $\mathcal{F}_w$ contains families with positive reference loss, $\mathcal{W}$ contains worlds with at least one such family, and vertical bars denote cardinality.

Event forecasting. EVENT averages Brier gain (Brier, 1950; Murphy, 1988) $G = \mathrm{Brier}(p_{\mathrm{ref}}, y) - \mathrm{Brier}(\hat{p}, y)$ for model probability $\hat{p}$, reference probability p_{ref}, and outcome y. Binary items use $\mathrm{Brier}(p, y) = (p - y)^2$ for event probability p and $y \in \{0, 1\}$, referenced to the supplied task-cell rate. Leader items use multiclass loss $\sum_j (p_j - \mathbb{1}[y = j])^2$, where p_j is the probability that civilization j leads and y the realized leader, with current-leader persistence as reference.

Hidden-state inference. TRAIT uses pairwise ranking accuracy, and RULES averages classification accuracy equally across rule parameters.

Intervention prediction. For cell c of quantity v, let $\hat{q}_{m,0.5}$ and y_m denote the predicted median and engine outcome under edit m. Subscript ctrl denotes the unedited control. Predicted and realized effects are $\hat{d}_c = [\varphi(\hat{q}_{m,0.5}) - \varphi(\hat{q}_{\mathrm{ctrl},0.5})]/\sigma_v$ and $d_c = [\varphi(y_m) - \varphi(y_{\mathrm{ctrl}})]/\sigma_v$. For an operator's eligible cells $\mathcal{C}$ with

nonzero realized effect d_c, Capture is

$$C = 1 - \frac{\sum_{c \in \mathcal{C}} |\hat{d}_c - d_c|}{\sum_{c \in \mathcal{C}} |d_c|}, \tag{3}$$

clipped to $[-1, 1]$ before averaging equally over operators with at least three eligible cells. No-effect and exact predictions score 0 and 1. Direction measures sign agreement among nonzero responses on these cells, counting missing answers as failures and weighting operators by eligible moved-cell counts.

Let $\bar{s}_{\text{real}}$ and $\bar{s}_{\text{pl}}$ denote mean $|\hat{d}_c|$ under real edits and matched placebos across valid cells and operators, including unchanged-truth cells. Valid-response Specificity is

$$\text{Spec}_{\text{valid}} = \text{clip}_{[-1,1]}\left(1 - \frac{\bar{s}_{\text{pl}}}{\bar{s}_{\text{real}}}\right), \tag{4}$$

defined when $\bar{s}_{\text{real}} > 0$ and both arms have valid shifts. It measures selectivity, not correctness, and should be interpreted alongside Capture.

Reference adjustment. Trait, Rules, and Direction report $S_{\text{adj}} = c(a - b)/(c - b)$ for raw accuracy a, baseline b, and reference ceiling c, which maps the baseline to zero without changing the ceiling or the model ordering. Missing or unparseable answers receive task-specific failure penalties, as specified in Section F.

4.2 Reference lines and composite index

Reference lines. FLOOR defines metric-specific origins, with Capture's clipped lower endpoint -1. EXPERT adds domain knowledge and a tool-assisted harness under the same observations, without hidden test state, future outcomes, or random seeds. The conceptual CEILING assumes complete simulator state and engine knowledge, but not future random draws.

Composite index. For model m, setting s, and metric k, let x_{msk} be the score, with floor f_{sk} and ceiling c_{sk}. Completion is $r_{msk} = (x_{msk} - f_{sk})/(c_{sk} - f_{sk})$, a fraction that tables report as a percentage. To define the Sandbox-World Forecasting Index (SWFI), let $\mathcal{S}_{\text{eval}} = \{\text{OMNI-ID}, \text{EGO}, \text{OMNI-OOD}\}$ and $\mathcal{K}_\ell$ contain Quantity and Event at level $\ell = 1$, Trait and Rules at $\ell = 2$, and Capture, Direction, and Specificity at $\ell = 3$. Then

$$\text{SWFI}(m) = \frac{100}{9} \sum_{s \in \mathcal{S}_{\text{eval}}} \sum_{\ell=1}^{3} \frac{1}{|\mathcal{K}_\ell|} \sum_{k \in \mathcal{K}_\ell} r_{msk}. \tag{5}$$

Metrics are weighted equally within levels, then across levels and settings. Weights and reference scales are fixed independently of external leaderboards, and completion and SWFI are not clipped.

4.3 Research questions

RQ1, validity. Do information controls, reference lines, and external forecasting associations support the evaluation? **RQ2, limits.** Where do current models fall short across task levels? **RQ3, process.** How does intermediate prediction quality relate to final answers, and how do reasoning protocols affect intervention prediction? **RQ4, utility.** Can world models improve agent decisions, and which capability profiles indicate utility?

4.4 Models and inference

We evaluate eleven language models with thinking enabled at model-specific effort settings recorded in Section M. Open-weight models, including Qwen3.8-Flash-Next, hereafter Flash-Next, cover the full grid. Claude Opus 5 and GPT-5.6 Sol, hereafter Opus and Sol, use medium reasoning effort on 190 shared frozen decision points, 10% of the benchmark, to bound API cost. This subset is stratified to match the full benchmark's composition by split, world size, budget, ruleset, decision time, and operator, one anchor per world, so subset scores estimate the same quantities as the full grid. EXPERT uses Sol on the same

anchors with a manual combining engine rules and regularities from 447 disjoint worlds. The Artificial Analysis (AA) Intelligence Index v4.3 (Artificial Analysis, 2026) provides a general-capability reference. The reported evaluations consumed about 3,600 H20-equivalent GPU-hours of open-weight inference, 5,000 CPU core-hours of engine simulation, and $2,500 of closed-model API calls.

5 Benchmark Validity and Model Capabilities

Table 3 summarizes the main results, with full setting-wise results in Table 14. We examine evaluation validity (RQ1), then identify model limitations and directions for improvement across task levels (RQ2).

5.1 Do the tasks and metrics support valid evaluation? (RQ1)

Insight 1: Information controls and engine checks support validity. Re-seeded rollouts in Section G.1 retain uncertainty despite complete state information, motivating distributional scoring. Figure 2 shows that removing history reduces forecasting performance, whereas study notes and privileged future narratives improve it, demonstrating evidence sensitivity. Construction checks in Section G.4 detect no leakage or consistency violations, and replay diagnostics in Section G.5 justify factual-rerun controls.

Insight 2: Reference lines locate the deficit in the models. Table 3 shows that EXPERT reaches 51% of CEILING, against 29% for the best model, so knowledge and harness support reveal substantial headroom under fixed observations, and the headroom lies in the models rather than in noisy targets. Branch noise lowers the Capture CEILING to $+0.39$ but cannot make a model negative, since predicting no effect scores 0. Nine of eleven models nonetheless score below -0.3 in OMNI-ID, while EXPERT reaches $+0.13$ and study notes raise Flash-Next by $+0.24$ on paired cells in Section H.3. Models shift forecasts as much under placebo as under real edits in Table 24, so their negative scores reflect over-reaction. For TRAIT the limit is the evidence: a privileged observer reaches only 0.59, so the task measures recovery of a weak but real signal, on which EXPERT and Sol still separate from models at chance.

Insight 3: Sandbox scores track real-world forecasting. Figure 3 shows positive correlations between SWFI and forecasting scores on ForecastBench (Karger et al., 2025), Metaculus FutureEval (Metaculus, 2026), and Prophet Arena (Prophet Arena, 2026), with $r = 0.79$–0.95 over panels of four to six matched models, and a model-level permutation test pooling the panels gives $r = 0.73$ with $p = 0.009$. These exceed SWFI's correlation of $r = 0.76$ with the AA general-capability index, and Table 13 shows the belief components nearly uncorrelated with AA. SWFI thus tracks forecasting-specific ability rather

Table 3: Model performance on CIVWORLDBENCH: three-setting SWFI and OMNI-ID diagnostics. Bold denotes the best model score, excluding the expert reference. Percentages indicate reference-normalized completion. Full three-setting results are reported in Table 14.

			L1 association		L2 belief		L3 intervention		
Model	AA	SWFI	Quantity ↑	Event ↑	Trait ↑	Rules ↑	Capture ↑	Direction ↑	Specificity ↑
FLOOR	—	0.00	+0.00	+0.000	0.00	0.00	−1.00	0.00	+0.00
gpt-oss-120b	12	−3.89	+0.03 (4%)	−0.031 (-29%)	0.01 (2%)	0.04 (5%)	−0.65 (25%)	0.15 (16%)	+0.05 (5%)
Gemma 4 31B	15	16.52	+0.21 (31%)	+0.012 (11%)	0.08 (12%)	0.11 (12%)	−0.61 (28%)	0.30 (33%)	**+0.19** (19%)
Nemotron 3 Ultra	23	−0.98	+0.08 (12%)	+0.003 (3%)	0.08 (11%)	0.12 (13%)	−0.63 (27%)	−0.16 (-17%)	−0.33 (-33%)
Inkling	26	15.73	+0.22 (32%)	+0.012 (11%)	0.13 (20%)	0.11 (12%)	−0.31 (49%)	0.09 (10%)	−0.11 (-11%)
Qwen3.8-27B	34	12.47	+0.25 (36%)	+0.007 (7%)	0.10 (16%)	−0.11 (-12%)	−0.34 (48%)	0.24 (26%)	−0.03 (-3%)
GPT-5.6 Sol	39	27.68	+0.27 (39%)	+0.022 (20%)	**0.23** (34%)	0.16 (17%)	−0.60 (29%)	0.28 (30%)	−0.02 (-2%)
DeepSeek-V4.1-Flash	40	20.48	+0.19 (27%)	+0.017 (16%)	0.14 (22%)	**0.17** (18%)	−0.58 (31%)	0.30 (32%)	+0.07 (7%)
Qwen3.8-Flash-Next	40	16.93	+0.26 (38%)	+0.011 (10%)	0.10 (15%)	0.08 (9%)	−0.57 (31%)	0.22 (24%)	+0.05 (5%)
Kimi K3	44	21.94	+0.34 (50%)	**+0.025** (23%)	−0.00 (-0%)	0.05 (6%)	−0.35 (47%)	0.23 (25%)	+0.08 (8%)
GLM-5.3	45	19.19	**+0.37** (53%)	+0.023 (21%)	−0.01 (-2%)	0.07 (8%)	−0.25 (54%)	0.16 (17%)	−0.05 (-5%)
Claude Opus 5	45	**29.47**	+0.33 (48%)	+0.021 (19%)	0.17 (26%)	0.13 (14%)	**−0.08** (66%)	**0.33** (35%)	+0.12 (12%)
EXPERT	—	51.18	+0.37 (55%)	+0.021 (20%)	0.26 (39%)	0.52 (56%)	+0.13 (81%)	0.58 (63%)	+0.86 (86%)
CEILING	—	100.00	+0.68	+0.108	0.66	0.93	+0.39	0.92	1.00

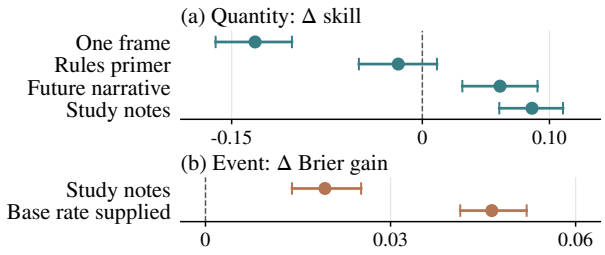


Figure 2: Flash-Next's OMNI-ID ablations with paired 95% world-bootstrap intervals. Table 17 and Section H.3 report scores and protocols.

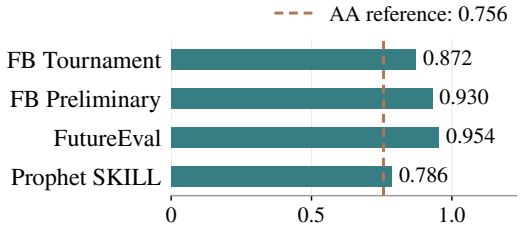


Figure 3: Pearson correlations between SWFI and real-world forecasting scores. Section G.3 provides details.

than general capability alone, which supports CIVWORLDBENCH as a proxy for real-world forecasting capability.

5.2 Where do current world models fall short? (RQ2)

Insight 1: Association forecasting strength does not reliably extend to belief inference and intervention prediction. Table 3 shows that GLM-5.3 leads the model panel in quantity forecasting, yet its trait score is below chance. Both GLM-5.3 and Kimi K3 lag behind Sol in belief inference and intervention direction. Such reversals recur across the panel, so association forecasting rankings do not consistently reflect belief and intervention capabilities.

The process analysis in Section 6 traces this gap to intermediate states, which models often mispredict even when their final answers are accurate and whose quality tracks intervention performance. The usability study in Section 7 further suggests that agent performance is more closely associated with L2 and L3 capabilities than with L1 forecasting alone, motivating evaluation beyond observation.

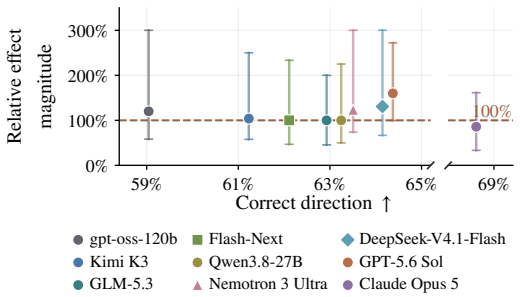


Figure 4: Direction accuracy and effect magnitude for nine models in OMNI-ID. Section H.4 adds the rest.

Insight 2: Effect prediction remains a shared weakness across models. Table 3 shows above-chance intervention direction for most models. Yet nearly all models have negative capture across settings, below the zero-effect baseline. Figure 4 shows informative direction predictions alongside widely dispersed effect magnitudes. Among correct-direction predictions, some medians approach 100%, while interquartile ranges span underestimation and overestimation. Section H.4 reports coverage and conditional results across settings, including 17–35% of correct-direction predictions in OMNI-ID, and up to 44% in OMNI-OOD, exceeding twice the realized magnitude. These errors arise from over-reaction to edits rather than from target noise. A single global rescaling cannot correct errors that fall on both sides of the truth, so effect magnitudes must be calibrated conditional on the world state.

6 Process Verification and Prediction Quality

Recorded trajectories and engine ground truth make intermediate predictions verifiable, linking process quality to intervention prediction and structured reasoning to process supervision (RQ3).

6.1 Tasks and experimental settings

We evaluate Qwen3.8-Flash-Next, Qwen3.8-27B, Gemma 4 31B, and Kimi K3 on 190 anchors, 10% of the benchmark, and GPT-5.6 Sol on a 30-anchor subset, each under three reasoning protocols. *Free* uses unrestricted reasoning without a prescribed process score. *Table* predicts prescribed intermediate states before answering, and *Code* generates a stochastic simulator, executed across 50 random seeds to yield intermediate predictions and final-answer distributions. Both are scored against engine truth at the same prescribed intermediate cells, with process quality defined as intermediate-state skill over the trend reference. Section I details the protocols.

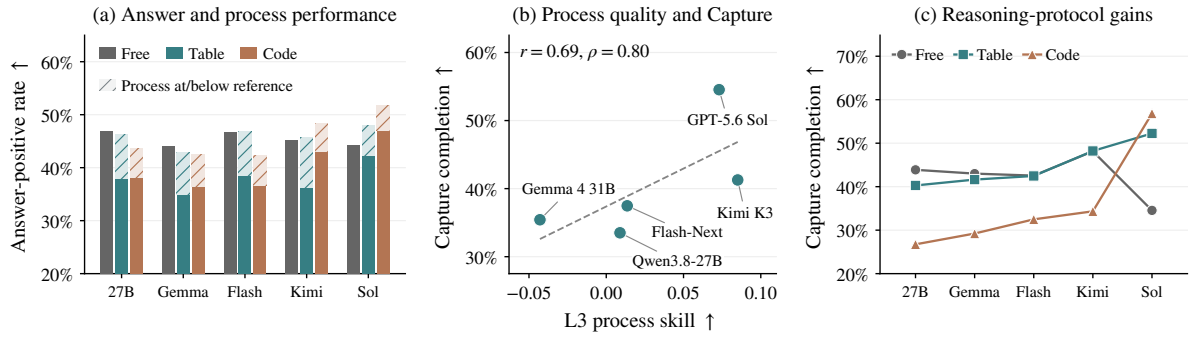


Figure 5: Reasoning protocols and prediction quality. (a) Above-reference answers, with hatching for intermediate forecasts at or below reference. (b) L3 process skill and Capture completion, averaged over Table and Code. (c) Capture completion by protocol. Scores average OMNI-ID and OMNI-OOD equally, and Section I details scoring and coverage.

6.2 How does process quality relate to prediction performance? (RQ3)

Insight 1: Strong final-answer performance can accompany weak intermediate forecasts. Figure 5a shows that 4.7–9.5% of matched predictions exceed the final-answer reference without exceeding the intermediate-state reference, across models and both structured protocols. Final-answer evaluation therefore misses intermediate prediction errors, motivating separate process verification.

Insight 2: Process quality is positively associated with L3 intervention prediction performance. Figure 5b shows that higher L3 process quality accompanies greater Capture completion across models. This association supports process verification as a complementary diagnostic and intermediate-state forecasting as a route to stronger intervention prediction. Section I.3 reports uncertainty.

Insight 3: Structured reasoning can improve intervention prediction. Figure 5c shows Table gains over Free increasing with SWFI. Sol benefits from both structured protocols and improves most with Code. Tables expose intermediate states for verification, while code connects them through executable dynamics. These results motivate world-model process supervision on states and intervention-induced changes, for which CIVWORLDBENCH provides a multi-agent counterpart to the verified executable worlds of Wang et al. (2026a).

7 World-Model Capabilities and Decision Utility

We assess whether world-model forecasts and advice improve agent decisions and how these gains relate to benchmark capability profiles (RQ4).

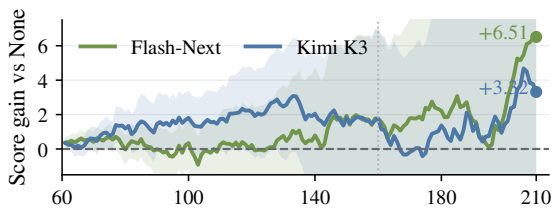


Figure 6: Mean gains over no WM in 76 worlds with pointwise 95% world-bootstrap intervals.

Table 4: OMNI-ID level completion and local gain over self-review. Bold indicates the highest completion. Table 32 reports uncertainty.

Model	L1	L2	L3	Δ score
Flash-Next	24.1%	12.3%	19.7%	+0.19
Kimi K3	**36.6**%	2.8%	26.7%	−0.66
Sol	29.7%	**26.0**%	19.0%	+1.40
Opus	33.5%	19.7%	**37.7**%	+1.28

7.1 Experimental setup and evaluation

A fixed Flash-Next agent plays turns 60–210 in 76 worlds, either without world-model assistance or with forecasts and advice from Flash-Next or Kimi K3 to inform plan revisions. A local study compares original, self-reviewed, and assisted plans at matched decision points in 36 validation worlds, using Flash-Next, Kimi, Sol, and Opus as advisers. Each plan undergoes four 20-turn executions without further strategic consultation. Both studies compare the engine's civilization score within worlds, as Section N details.

7.2 Do better world models support better decisions? (RQ4)

Insight 1: World models can support agent decision-making. Figure 6 traces score gains over turns 60–210, with positive mean gains for both assisted conditions at turn 210. In the local study, selecting plans on trial executions and evaluating them on held-out executions, adding WM-generated plans improves selection by 2.36 score points over choosing among original and self-reviewed plans only, so advice expands the set of useful alternatives. The case studies show what useful advice consists of: Opus corrects the agent's reading of a conditional loss probability as a certainty and redirects its remaining gold to one fortified defender, saving two cities, whereas Sol's precautionary tax increase buys defenses against a threat that never materializes and costs research. Useful advice therefore hinges on reading the opponent's intentions and the state-specific cost of a precaution, which are belief and intervention judgments rather than trend extrapolation. Figures 13 and 14 illustrate how specific recommendations change decisions.

Insight 2: Decision-oriented evaluation should extend beyond association forecasting. Table 4 shows larger estimated decision gains for Opus and Sol than Kimi despite lower L1 scores, and these two advisers also lead the panel in belief inference and intervention direction. Decision utility thus tracks the capabilities that association forecasting leaves unmeasured. This motivates evaluating hidden-state inference and action-conditioned effect prediction alongside association forecasting.

8 Conclusion

CIVWORLDBENCH evaluates forecasting, hidden-state inference, and intervention prediction in multi-agent worlds with engine-verifiable targets. Its composite score tracks real-world forecasting more closely than general capability, while level diagnostics expose weaknesses hidden by observational accuracy: intermediate-state errors, poorly calibrated effects, and advice utility that forecasting alone does not explain. These findings prioritize state representation, effect calibration, and verifiable process supervision. Taken together, the results characterize current language models as competent trend extrapolators but weak causal simulators of multi-agent worlds, and the benchmark supplies the hidden-state and intervention targets needed to train the latter. Section B discusses limitations.

AI use statement

Generative AI tools, including Claude and Codex, assisted implementation and manuscript polishing, and were used for retrieval and discovery, such as finding related work and checking bibliographic details, which the authors then verified against the original sources. Language models also produced the forecasts, reasoning, executable predictors, and advice evaluated in the experiments. These outputs are distinct from the ground-truth states and outcomes generated by the game engine. Expert-harness and mechanism-card uses are documented with their experimental protocols. The authors are responsible for the implementation, analyses, and manuscript writing.

Ethics statement

The study involves no human participants. Benchmark trajectories use engine-controlled agents, and the usability study introduces language-model-controlled decisions within the same simulated environment. Freeciv is distributed under the GPL, whose requirements apply to engine modifications. Benchmark performance does not validate policy recommendations for human societies. The icons used in our figures are sourced from Flaticon[2] and are free for personal and commercial purposes with attribution.

Reproducibility statement

Scoring and analysis code, engine patches, and build instructions are available at `https://github.com/Leon-Francis/CivWorldBench`. The benchmark data release, including frozen questions, targets, and model predictions, is in preparation. Section M records evaluation configurations and result provenance. Section G specifies reference-line estimation, and Section D documents the collection pipeline and validation checks.

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A Related Work

World-model evaluation and intervention prediction. WorldPrediction evaluates action and action-sequence selection from video states (Chen et al., 2025), while Li et al. (2026) survey text world models for agents. Corr2Cause tests causal inference from statistical descriptions (Jin et al., 2024), while CausaLab supports interactive causal discovery in synthetic laboratories (Yang et al., 2026a). These studies examine dynamics prediction and mechanism recovery under different task formulations. CIVWORLDBENCH evaluates forecasting, hidden-state inference, and intervention prediction within the same multi-agent dynamics. Recorded trait and rule values provide inference targets, while paired engine rollouts measure intervention outcomes under controlled observations.

Real-world and social forecasting. ForecastBench evaluates forecasts of future events (Karger et al., 2025). Prophet Arena separates event collection, prediction, and evaluation in a live forecasting pipeline (Yang et al., 2026b), while Lu (2025) compares model forecasts with expert predictions on Metaculus questions. SocietyBench evaluates social-event forecasting on anonymized, date-shifted timelines (Wang et al., 2026b), and SWM-Bench uses prediction-market trajectories to evaluate collective belief transitions (Yu et al., 2026). CIVWORLDBENCH complements these outcome-based evaluations with recorded hidden states and repeated stochastic futures. Section G.3 relates its composite scores to real-world forecasting through matched-model comparisons.

Multi-agent environments and decision utility. SOTOPIA evaluates social interaction (Zhou et al., 2024), while social world models explicitly represent other agents' mental states (Zhou et al., 2025). CivRealm and BALROG assess decision-making agents through game interaction (Qi et al., 2024; Paglieri et al., 2025). End-to-end performance combines dynamics knowledge, planning, and action execution. CIVWORLDBENCH separately evaluates an observer of engine-controlled civilizations, then holds the acting model fixed when comparing world-model advisers. This design connects prediction diagnostics to decision utility while controlling the agent's execution capabilities.

Process verification and executable worlds. Rahmani (2026) distinguishes action-generation errors from state-propagation errors in code world models. Verifier-driven executable models also support planning in ARC-AGI-3 by checking simulated transitions against previous observations (Rodionov, 2026). Code as Worlds uses verified executable worlds to derive physical-reasoning supervision (Wang et al., 2026a). CIVWORLDBENCH extends process evaluation to stochastic multi-agent trajectories, checking Table and Code predictions at shared intermediate states and relating their quality to final intervention forecasts. These targets support studying supervision for both state prediction and intervention-induced changes.

B Limitations

Scope. All dynamics come from one engine and its built-in AI. The generalization set changes the ruleset, difficulty, map, civilization count, hidden-parameter ranges, and team structure, but remains within that engine. The benchmark does not test transfer to other simulators or to human societies.

References and targets. The Trait and Direction ceilings are computed on identifiable subsets, namely trait gaps of at least 60 and cells with $|d_c| > \sigma_c$, so completion in these columns is conservative, and the Capture ceiling in Section G.2 assumes Gaussian branch divergence. Intervention targets include the reload-path divergence measured in Section G.5. This divergence bounds attainable Capture but cannot produce negative scores, which Section 5 attributes to the models. Traits are only weakly identifiable from 80 turns of behavior, even for a privileged observer, whose accuracy is 0.59, so TRAIT scores should be read against that observer. EXPERT is a language model with a manual and tools, not a human baseline.

External and downstream evidence. The matched forecasting panels contain four to six models, and a general-capability index is also correlated with the same leaderboards, so proxy validity rests on the component-level evidence in Table 13 as much as on the composite. The process study covers five models, and agent play uses one actor in benchmark worlds. Their model-level association and full-game gain intervals include zero, as Sections I and N report.

C Formal definition

World and observation. In the process $(\mathcal{S}, \mathcal{A}, T, \Omega, O_v)$ of Section 2, a world $w \sim \mathcal{W}$ specifies a ruleset, map, agent population, difficulty, hidden parameters, and team layout. The action set $\mathcal{A}$ collects the engine commands available to an agent, and Ω is the space of rendered reports. Determinism conditional on an engine random stream does not imply that an edited reload consumes that stream identically to an unedited run. The rendered policies $\{\pi_i\}$ describe current aims rather than future commitments.

Queries. The world model $f(o_{\le t_1}, \{\pi_i\})$ answers four query types. With g a visible quantity of one agent, e an event predicate on a trajectory segment, and m an operator on states, they are

$$\begin{aligned}
\textsc{Quantity}: &\quad P(g(s_{t_1+h}) \mid o_{\le t_1}, \{\pi_i\}) \text{ reported as five quantiles,}\\
\textsc{Event}: &\quad P(e(s_{t_1:t_1+h}) = 1 \mid o_{\le t_1}, \{\pi_i\}) \text{ for binary items,}\\
\textsc{Trait, Rules}: &\quad P(\theta_k \mid o_{\le t_1}) \text{ as a ranking or a bin, policies withheld,}\\
\textsc{Intervene}: &\quad P(g(s'_{t_1+h}) \mid o_{\le t_1}, \{\pi_i\}, \operatorname{do}(s_{t_1} \leftarrow m(s_{t_1}))),
\end{aligned}$$

where s' follows T from $m(s_{t_1})$ under the unchanged policies, and θ_k denotes a queried trait or world parameter in θ_w. Mainline trajectories provide forecast targets, recorded hidden fields provide belief targets, and rerun arms provide intervention targets. Realized branch differences are not noise-free conditional average treatment effects. Matched placebos measure response to negligible or mechanism-irrelevant edits, while the model's own no-intervention forecast defines its predicted effect.

References. A reference predictor is a fixed map $r(o_{\le t_1})$. Persistence uses the current value, while the trend reference extrapolates the two most recent frames. The conceptual ceiling knows T and s_{t_1} but not future u. Re-seeded reruns approximate its forecasting loss. The information sets of the ceiling and expert references are

$$\hat{y}_{\text{ceil}} \in \arg\min_{\hat{y}} \mathbb{E}[\ell(\hat{y}, Y) \mid s_{t_1}, T],$$

$$\hat{y}_{\text{expert}} = h_{\mathcal{H}}(o_{\le t_1}; \mathcal{K}),$$

where $\mathcal{K}$ denotes expert knowledge and $\mathcal{H}$ the harness. The ceiling has complete state information, while the expert retains the model's rendered history and adds knowledge and computation. The process evidence in Section 6 consists of claims about $g(s_\tau)$, $t_1 < \tau < t_1 + h$, each scored against the trajectory.

D Corpus construction and engine patches

D.1 World generation

The evaluation corpus contains 475 worlds with four nominal decision points per world. The basic set fixes the ruleset to `civ2civ3` and difficulty to hard, with 75 worlds in each of three map and population classes. The generalization set jointly varies ruleset, difficulty, map, civilization count, hidden parameters, and team structure. Table 1 summarizes the map, population, and budget ranges. Hidden-parameter ranges are wider than in the development pool. The resulting shift does not require every OOD world to be harder than every ID world.

Mainline collection saves every turn and runs beyond the reporting budget to provide future targets. The reporting budgets are 400, 800, and 1,200 turns. Decision points are distributed from turn 60 to the

Table 5: Source patches used to construct the benchmark engine.

Patch	Modification
Trait initialization	Corrects trait arithmetic and initializes diplomatic contact at peace.
Diplomatic thresholds	Revises ceasefire, peace, and war decision thresholds.
Alliance dynamics	Reduces persistent alliance-bloc warfare.
Aggression response	Connects aggression to ceasefire acceptance and war persistence.
Production preference	Connects building preference to the active production decision.

reporting endpoint. Budget-specific forecast horizons and the final post-filtering prompt inventory are defined separately in Section E.

D.2 Engine and ruleset modifications

The benchmark uses Freeciv 3.1.6 with five source patches that make selected behavioral parameters effective and permit changes in diplomatic relations. Table 5 summarizes their roles. Ruleset modifications widen trait ranges and enable domestic and intercontinental trade. Illness variants alter generation but their suffixes are not shown in model observations. The corpus and all intervention arms were collected with the resulting image, tagged `freeciv-server:3.1.6-v2`. The optional alliance-threshold patch is unapplied, and the later task-persistence patch enters only the agent-play image of Section N, not the benchmark engine.

Development checks tested diplomatic transitions, trait sensitivity, rule identifiability, team-game completion, and intervention persistence before model evaluation. They establish behavioral variation, while the final evidence gates define which inference questions are asked.

D.3 Replacements and branch collection

Three worlds were replaced after crashes or premature termination prevented target collection. Only their replacements enter the final question set.

Figure 7 shows the four branches collected from each eligible checkpoint. Collection produces an unedited factual rerun, a factual recheck, a real intervention, and a matched placebo. Branches are saved every turn. Validation checks the applied edit, save continuity, and factual/recheck agreement over comparable saved states. Successful short runs contribute only horizons with recorded truth. Failed engine states are retried under recorded settings, including anchor-level reseeding where needed, and their control arms are regenerated consistently. The branch metadata records these deviations. No model prediction is used to select a retry.

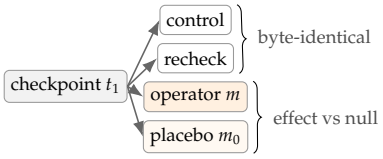


Figure 7: Four branches from a shared checkpoint provide replay checks and intervention controls.

Matching factual and recheck digests verifies replay, not shared randomness after an edit. End-turn settings, save frequency, reload behavior, and interventions can change subsequent random-number consumption. Section G.5 reports placebo and reseeding diagnostics.

D.4 Integrity and provenance

Each world and branch record identifies the engine configuration, initial state, intervention, observation window, and available targets. The question verifier checks scoped leakage and consistency constraints before scoring. Section G.4 reports these checks. Versioned checksums distinguish changes to experimental inputs from changes to scoring or presentation.

E Question construction

E.1 Observation windows and question scopes

Each question is associated with a world, decision turn, task, and observation condition. With the frame spacing of Section 2, an anchor at turn 60 uses turns 20, 40, and 60. A narrative summarizes intervening events without including events after the anchor. The egocentric viewer is the median-scoring living civilization at that decision point. Its question scope includes itself and living civilizations it has contacted.

Belief questions omit the rendered policies because policy fields can reveal their hidden targets, and truth fields remain outside the model-facing prompt.

E.2 Numerical forecasts

QUANTITY requests the 10th, 25th, 50th, 75th, and 90th percentiles of each eligible quantity. Treasury, population, city count, total city size, military and civilian unit counts, and enemy and ally counts are levels. Score, culture, and technology targets are increments from the decision point. Repeated Future Technology advances do not increase the technology count. Forecast horizons are 60/120, 120/250, and 200/300 turns for reporting budgets of 400, 800, and 1,200 turns, respectively.

Unavailable engine targets are not imputed. Constant culture in classic-rule worlds and technology increments after all technologies are known are excluded by the scorer's eligibility rules. These exclusions are distinct from missing model answers, which receive the penalties specified in Section F.

E.3 Events and supplied base rates

Binary templates describe occurrences within a horizon or comparisons between the initial and final state. Leader items instead request a categorical distribution over civilizations. The first filter retains binary template–budget–horizon–decision-turn cells with empirical rates between 0.1 and 0.9. A second pass checks the reference cells actually used by each evaluation setting. Cells with at least 200 items use the setting-specific empirical rate. Sparse cells fall back to the corresponding corpus cell, which must also contain at least 200 items and satisfy the same rate filter. The final reference is therefore determined before model scoring, not by its valid outputs.

Each retained binary question prints its reference rate. Leader items retain their original categorical format and current-leader persistence reference. Intervention event prompts use the same reference-cell machinery. These rates are in-sample benchmark statistics supplied equally to all models, not estimates from an independent training set.

E.4 Hidden traits and world rules

Trait questions rank civilizations separately on aggression, expansion, trade, and building preference. Each ranking includes every rankable civilization exactly once and permits predicted ties. True ties are excluded from pairwise scoring. A scope with fewer than two rankable civilizations has no trait-ranking question.

World-rule questions classify disaster frequency, growth cost, and research cost into three bins. Disaster bins are at most 12, 13–40, and above 40. Growth- and research-cost bins are at most 82, 83–117, and above 117. Lower costs correspond to faster growth or research. Growth questions require at least one observed city-size change, and research questions require at least two completed technologies within the observation window. Egocentric evidence gates use the viewer's evidence rather than inaccessible foreign events. Disaster questions have no additional event-count gate. Illness is not a benchmark target.

Table 6: Benchmark intervention families and numerical scoring targets. Bands are in turns.

Intervention	Scored quantities	Band
Treasury change	Treasury	5–30
Research points	Technologies learned	5–60
City destruction	Total city size, population	5–150
Government-enabling technology	Score increment	5–60
Map reveal	City count, score increment	30–300
Aggression trait	Enemy count, military units	50–300
Expansion trait	City count, total city size	50–300
Trade trait	Treasury, technologies learned	50–300
Building trait	Total city size, military units	50–300
Attitude change	Enemy and ally counts	50–300
Disaster frequency	Population, total city size	30–300

E.5 Intervention prompts and eligibility

An intervention prompt retains the history and describes the edit. Visible-state edits also update the affected current observation. Numerical and event subquestions are separate prompts, and a no-intervention control prompt is added when the required horizon or question set differs from the ordinary forecast. Each intervention is paired with this explicit control or the ordinary forecast, matching the target variables and horizons.

Table 6 lists the numerical scoring targets. Standard budget horizons are used when they fall inside an operator's declared band. Otherwise the longest resolved arm horizon in that band is selected. The available arm horizons are 30, 60, 90, 120, 150, 200, 250, and 300 turns. The target mask is fixed before inspecting a model's errors.

Real and placebo edits share their operator family but differ in magnitude or mechanism. For example, city-destruction arms remove large versus small cities. City count is excluded from their numerical target mask because both edits remove the same number of cities. Likewise, the government-enabling and placebo technology grants each add one technology, so their numerical target is a downstream score change rather than the identical count increment. Four single-trait operators replace the earlier composite-trait edit. Their collected modifiers are +90 for real edits and +3 for placebos. An accepted edit need not produce a favorable or nonzero downstream effect in every state. Main capture uses truth-moved cells, whereas specificity retains both moved and unmoved response cells.

F Metrics

Quantity aggregation. The headline QUANTITY score pools model and reference losses within each world and quantity family before taking the ratio in Equation (2). Families with zero reference loss are omitted. Valid families are averaged within each world, followed by equal weighting of worlds. Enemy and ally counts use the absolute error of the predicted median rather than the five-quantile loss in Equation (1). The appendix's family breakdown additionally equalizes variables within each family before averaging worlds. It is a diagnostic and its columns need not average to the headline QUANTITY score. The constants and variable transforms are shared across compared conditions. Pooling reduces sensitivity to small per-cell denominators but does not eliminate sensitivity when aggregate reference loss is small.

Events and hidden attributes. Event gain combines calibration and discrimination, without identifying them separately. The template plot uses a Brier *skill ratio* only as a diagnostic. TRAIT averages pairwise ordering accuracy over four traits. Predicted ties receive half credit and true ties are excluded. RULES averages accuracies over the three rule parameters equally. Its majority reference and observer reference use the same parameter weights, rather than weights proportional to question counts.

Table 7: Underlying accuracies before the main table's reference adjustment. Triples are OMNI-ID/EGO/OMNI-OOD. Raw references are 0.50 for trait and direction, and the equal-parameter majority accuracy for rules.

Line	Trait accuracy	Rule accuracy	Direction accuracy
Raw reference	0.50 / 0.50 / 0.50	0.40 / 0.40 / 0.41	0.50 / 0.50 / 0.50
gpt-oss-120b	0.50 / 0.49 / 0.52	0.43 / 0.44 / 0.46	0.57 / 0.56 / 0.54
Gemma 4 31B	0.52 / 0.51 / 0.54	0.47 / 0.46 / 0.46	0.64 / 0.63 / 0.59
Nemotron 3 Ultra	0.52 / 0.50 / 0.51	0.47 / 0.46 / 0.40	0.43 / 0.42 / 0.41
Inkling	0.53 / 0.51 / 0.55	0.47 / 0.48 / 0.44	0.54 / 0.56 / 0.48
Qwen3.8-27B	0.52 / 0.51 / 0.54	0.34 / 0.30 / 0.29	0.61 / 0.61 / 0.57
GPT-5.6 Sol	0.56 / 0.52 / 0.57	0.50 / 0.48 / 0.51	0.63 / 0.65 / 0.69
DeepSeek-V4.1-Flash	0.53 / 0.51 / 0.56	0.50 / 0.47 / 0.51	0.64 / 0.60 / 0.59
Qwen3.8-Flash-Next	0.52 / 0.51 / 0.54	0.45 / 0.47 / 0.39	0.60 / 0.62 / 0.58
Kimi K3	0.50 / 0.50 / 0.52	0.43 / 0.45 / 0.48	0.61 / 0.61 / 0.61
GLM-5.3	0.50 / 0.50 / 0.49	0.44 / 0.46 / 0.43	0.57 / 0.60 / 0.55
Claude Opus 5	0.54 / 0.51 / 0.56	0.48 / 0.48 / 0.51	0.65 / 0.72 / 0.59
EXPERT	0.56 / 0.53 / 0.57	0.70 / 0.61 / 0.71	0.76 / 0.79 / 0.65

Baselines for the reference adjustment. The adjustment in Section 4 uses $b = 0.5$ for trait and direction accuracy and, for rules, the equal-parameter majority accuracy on the complete truth set, independent of a model's missing or unparseable answers. Table 7 reports the underlying accuracies.

Missing or unparseable outputs. We retain required answer cells that a model omits or renders unparseably and count them as failed predictions, using the same policy for every model. For QUANTITY, a missing cell contributes twice its reference loss to the model-loss accumulator before the existing family aggregation and clipping. A wholly missing family therefore receives skill -1. The same rule uses the MAE reference for diplomacy. Families with zero reference loss remain undefined. Missing binary event probabilities receive Brier loss 1, and missing leader distributions receive multiclass Brier loss 2. Missing trait orderings and categorical labels receive zero accuracy. These penalties do not impute a forecast. For intervention capture, a missing arm or control prediction in cell c contributes $2|d_c|$ to the effect-error numerator, and its direction is counted as incorrect. For specificity, let r be the fraction of required response cells with valid arm and control predictions. We report $r\,\mathrm{Spec}_{\mathrm{valid}} - (1 - r)$, with score -1 when no responses are valid. Here $\mathrm{Spec}_{\mathrm{valid}}$ is the valid-response selectivity score in Equation (4). An all-valid set retains the original statistic. Raw loss and calibration diagnostics still require actual predictions. Incomplete experiments are reported at their available evaluation scope. Entirely unrun anchors are not counted as failed model attempts.

Interventions. Capture uses truth-moved cells, $d_c \neq 0$, of the target civilization within each operator's declared variable set and horizon band, applying the failure convention above. Zero predicted responses are excluded from the direction denominator. Correctness and selectivity use different denominators, so a specificity of 1 can accompany inaccurate real-arm predictions, and negative capture can accompany above-chance direction accuracy. For reference ceiling c_{Capture}, the completion percentage is $100(C + 1)/(c_{\mathrm{Capture}} + 1)$, not the improvement over a no-effect prediction, which attains $C = 0$ inside this display interval.

Paired analyses and uncertainty. The paired analyses in Section 5 intersect valid cells within each comparison and retain the original aggregation, truth, and reference losses. A cell is indexed by world, decision point, operator and arm when relevant, civilization, target, and horizon. Repeated event items with the same scored key retain their source-order occurrence index, assigned before validity filtering. The ablation builders preserve this order. Paired cells must have identical ground truth. Missing predictions are excluded jointly, so the paired estimate is conditional on common valid outputs and is not a missingness-adjusted full-run estimate. QUANTITY and EVENT intervals resample worlds jointly

between conditions with 1,000 replicates and seed 2026. For capture, exponential world weights give a paired Bayesian bootstrap that keeps the eligible operator set fixed. If any included operator has only one world, no between-world interval is reported. These intervals do not quantify variation across repeated model calls, and the exploratory comparisons are not adjusted for multiple testing.

G Validation, reference lines, and evaluation noise

G.1 Repeated-future reference protocol

We sample 92 decision points stratified by evaluation set and turn budget. Each contributes 8 re-seeded reruns saved every turn, combined with a historical factual rerun to form the nine-member reference ensemble. The mainline's rank among the nine trajectories has mean probability integral transform 0.49, compared with 0.5 under exchangeability. Fair CRPS and fair Brier losses correct for finite ensemble size. We report comparisons against mainline truth, leave-one-out comparisons against rerun truth, and an ensemble evaluated with the benchmark scorer. Leave-one-out estimates are the primary event reference because reload behavior can change the event distribution relative to the uninterrupted mainline.

Reload triggers AI government re-evaluation, changing the government-event distribution. Ten-turn saves also miss brief treasury-to-zero events, shifting their reference from -0.43 to +0.29 with per-turn recording. The historical factual member retains ten-turn evaluation, leaving a residual resolution mismatch. These template-specific distribution shifts remain despite the near-central aggregate rank. In EGO, the full-checkpoint observer remains a privileged reference rather than the Bayes limit of the rendered history.

G.2 Reference definitions and interpretation

G.2.1 Information sets of the numerical ceiling

Repeated futures approximate the forecast loss of the conceptual CEILING, while the other columns use privileged observers, noise models, or an algebraic endpoint. Trait and rule observers infer labels from privileged behavioral evidence with target fields withheld, so their scores characterize empirical recoverability rather than proven information limits.

G.2.2 Trait and rule references

The trait observer is leave-world-out ridge regression on 33 privileged behavior statistics from the 80 turns preceding the decision point. Its accuracy ranges from 0.56–0.62 across traits, averaging 0.59 over all unequal pairs. Restricting true trait gaps to at least 40 raises the four-trait mean to 0.63, with a per-trait range of 0.59–0.66, and gaps of at least 60 raise it to 0.66. The displayed 0.66 is this well-separated-pair reference, whereas the model scores include all eligible unequal pairs. The accuracies in this paragraph precede the main table's reference adjustment. The improvement with larger trait gaps indicates that behavioral evidence is more informative for strongly separated dispositions. The rule observer inverts growth and research-cost formulas where evidence is available and uses a leave-world-out classifier on disaster counts per city-turn for the remaining parameter. Disaster accuracy is 0.79 / 0.59 / 0.72. Averaging the three parameter references equally yields 0.93 / 0.86 / 0.90.

G.2.3 Intervention noise model

Let d_c be the observed real-arm effect and σ_c the root-mean-square placebo effect for the corresponding operator and quantity. The direction reference uses $\Phi(\sqrt{\max(d_c^2 - \sigma_c^2, 0)}/\sigma_c)$, where Φ is the standard normal cumulative distribution function. It is 0.71 / 0.71 / 0.68 over all moved cells and about 0.92 on the 48% (basic) / 42% (generalization) of moved cells with $|d_c| > \sigma_c$, whereas models are scored on all moved cells. The capture reference assumes centered Gaussian divergence and exact prediction of

Table 8: Validation evidence and reference assumptions. Triples denote OMNI-ID/EGO/OMNI-OOD. Numerical references differ in information and estimation procedure.

Check	Observation	Interpretation
Repeated futures	QUANTITY $\simeq +0.68$, EVENT $\simeq +0.11$	Predictable signal remains under engine randomness with full-checkpoint information.
Trait observer	0.59 on all pairs, 0.63 at gap $\geq$40, 0.66 at gap $\geq$60	Privileged behavioral features provide partial recovery of trait orderings.
Rule observer	0.93 / 0.86 / 0.90	Formula inversion and disaster statistics support parameter-specific identifiability.
Direction reference	all moved cells 0.71 / 0.71 / 0.68	The displayed 0.92 concerns only effects above estimated placebo divergence.
Capture reference	+0.39 / +0.39 / +0.30	A reference derived under a Gaussian noise decomposition.
Specificity endpoint	1.00	An algebraic endpoint for real-only responses.
Expert reference	QUANTITY +0.37 / +0.29 / +0.37	A positive control on 190 anchors with additional knowledge and harness support.

the underlying mechanism, so expected absolute residual error is $\sigma_c\sqrt{2/\pi}$. Substituting this into the operator-level capture ratio gives $1 - \sum_c \sigma_c\sqrt{2/\pi} / \sum_c |d_c|$, then the operator macro-average +0.39 / +0.39 / +0.30. Cell pooling gives +0.29 / +0.29 / +0.20, illustrating the effect of operator weighting. These estimates assume Gaussian divergence and transferable placebo variance. Completion expresses progress along the resulting reference scale.

G.2.4 Construction of the expert reference

The published EXPERT reference combines GPT-5.6 Sol with the frozen v3.3 manual and a tool-assisted harness on the shared 190-anchor sample.

Additional expert knowledge. The v3.3 manual summarizes game rules, default-AI logic, engine patches, and empirical regularities from non-benchmark worlds. It covers growth and research formulas, behavioral cues for hidden targets, forecast intervals, and intervention reasoning, without target-world answers or model scores.

Harness and information boundary. An independent agent handles each anchor–condition pair using the manual, canonical observations, task prompts, and a fixed answer inventory. It maintains working notes, uses Python for calculations, and answers sequentially to coordinate factual and intervention predictions. Access is restricted to this input pack, excluding external data, prior predictions, scores, hidden state, and future outcomes. The harness adds computation and multi-step reasoning without changing the evaluated world's information boundary.

Validation and interpretation. Table 14 reports the combined effect of expert knowledge, tools, and reasoning resources under the same observations. Evaluation uses the fixed v3.3 manual and input packs, with answer inventories checked before scoring and standard penalties for missing or unparseable cells. Later manual revisions and expanded-information pilots are excluded.

G.3 Association with real-world forecasting performance

External associations. Table 9 shows positive Pearson and Spearman correlations between SWFI and matched-model forecasting scores across all reported tracks. Agreement in both score variation and ranking supports the use of CIVWORLDBENCH as a proxy for real-world forecasting capability.

Evaluation scope. All comparisons use the same scores, missing-answer penalties, and reference scales as Tables 3 and 14. The analysis uses 1,900 anchors for open models and 190 for Opus 5 and GPT-5.6 Sol.

Table 9: Associations between fixed-weight SWFI and external model scores. n counts matched models. Ranges span eligible external configurations. The two ForecastBench tracks overlap. AA provides a general-capability reference. Section G.3 describes the protocols and sensitivity analyses.

External reference	n	Pearson r	Spearman ρ
ForecastBench Tournament (Overall)	4	0.864–0.879	0.800
ForecastBench Preliminary (Dataset)	5	0.930	0.900
Metaculus FutureEval (Avg Score)	5	0.954	0.975
Prophet Fixed Context (SKILL)	6	0.786	0.886
AA Intelligence Index v4.3 (reference)	11	0.756	0.795

Table 10: SWFI and external scores for exact model matches. T/P denotes ForecastBench Tournament/Preliminary and FE denotes FutureEval. SKILL uses Prophet's Fixed Context track. Dashes indicate unavailable matches. External scores retain their native units.

Model	SWFI	FB T	FB P	FE	SKILL	AA
GPT-6 Astra	36.59	–	–	–	-0.98	–
Claude Opus 5	29.47	66.10–66.30	61.70	15.78	–	45
Gemini 3.7 Flash	28.75	–	62.00	–	-0.91	–
GPT-5.6 Sol	27.68	65.40–65.50	61.50	12.93	-1.05	39
GLM-5.3 Flash	22.57	–	–	–	–	–
Kimi K3	21.94	65.60	60.90	12.93	-1.14	44
DeepSeek-V4.1-Flash	20.48	–	–	–	–	40
GLM-5.3	19.19	–	–	–	-1.08	45
Qwen3.8-Flash-Next	16.93	–	–	–	–	40
Gemma 4 31B	16.52	61.50	57.40	–	–	15
Hunyuan Hy3	15.83	–	–	–	-1.17	–
Inkling	15.73	–	–	–	–	26
Qwen3.8-27B	12.47	–	–	–	–	34
Nemotron 3 Ultra	-0.98	–	–	5.83	–	23
gpt-oss-120b	-3.89	–	–	-0.26	–	12

Gemini 3.7 Flash, GPT-6 Astra, GLM-5.3 Flash, and Hunyuan Hy3 add evaluations on the same 60 anchors under the same scoring policy. Both subsets are stratified to match the full benchmark's composition, as described in Section 4, so the three sample sizes estimate the same quantities at different precision. As a check, SWFI recomputed on the 60 anchors is 27.5 for Sol against 27.7 on its 190 anchors, and 25.1 for Kimi K3 against 21.9 on the full grid. Neither difference changes either model's rank among the matched models on any panel. The assisted expert is excluded as a non-independent model.

Expansion and matching protocol. Weights and scales were fixed before evaluating the four additional models, selected for external overlap and API access. Leaderboard eligibility was refined during exploratory analysis after inspecting existing results. Table 10 lists exact model identities, local scores, and eligible matches. Each model contributes one observation, with multiple external configurations handled as described below.

External sources and configurations. ForecastBench (Karger et al., 2025; ForecastBench, 2026) uses the September 22, 2026 organizer tables for Tournament Overall and Preliminary Dataset. Preliminary remains provisional, and the two tracks overlap. Eligible rows retain their search/code and reasoning configurations. For multiple eligible variants, Table 9 reports all correlation combinations. The main-text figure correlates each model's median external score across configurations.

Metaculus FutureEval (Metaculus, 2026) uses the archived leaderboard PDF dated September 23, 2026 and its log-score-based Avg Score. All five matches use AskNews variants, although Opus and GPT reasoning levels differ from the local medium settings. Matching distinguishes Gemma 26B from Gemma 31B.

Prophet Arena (Prophet Arena, 2026) uses the Fixed Context SKILL snapshot retrieved September 23,

Table 11: External-validity sensitivity. Drop-one ranges omit each model in turn and include all eligible configurations. Exact two-sided permutation tests are exploratory and unadjusted.

External reference	Drop-one r	Permutation p
ForecastBench Tournament (Overall)	0.455–1.000	0.0833–0.1250
ForecastBench Preliminary (Dataset)	0.917–0.994	0.0250
Metaculus FutureEval (Avg Score)	0.937–0.987	0.0167
Prophet Fixed Context (SKILL)	0.673–0.913	0.0667

Table 12: Same-model comparison with AA: each predictor is correlated with the external outcome. This differs from the SWFI–AA association in Figure 3. Only panels with at least three recorded AA values are included.

External reference	n	SWFI r	AA r
ForecastBench Tournament (Overall)	4	0.864–0.879	0.991–0.994
ForecastBench Preliminary (Dataset)	4	0.917	0.964
Metaculus FutureEval (Avg Score)	5	0.954	0.987
Prophet Fixed Context (SKILL)	3	0.510	-0.645

2026 at 16:17 China Standard Time, with six exact model matches. SKILL is the fitted market-anchored parameter θ. Retired models retain their last scores. Evidence and inference budgets can differ within the shared track. FutureX is excluded because a common-framework comparison could not be established.

Sensitivity and uncertainty. Pearson correlation measures linear association, while Spearman correlation measures rank agreement. Each comparison uses models with observed scores on both benchmarks.

Table 11 shows that Preliminary and FutureEval retain strong correlations after each model is removed, while Tournament is more sensitive to model removal. The reported ranges cover leave-one-model-out estimates and eligible configurations. Exact two-sided permutation tests are exploratory and unadjusted for multiple comparisons. With four matched models the smallest attainable two-sided p is 0.083, so the Tournament panel cannot reach significance at any correlation. The pooled test in Section 5 standardizes each panel's external scores, pools the 20 model–panel pairs over 10 distinct models, and permutes SWFI across models so that a model's pairs move together, with 200,000 draws. It gives $r = 0.73$ and $p = 0.009$, and $r = 0.73$ with $p = 0.006$ after dropping the Tournament panel, which overlaps Preliminary. The SWFI–AA correlation over eleven models has permutation $p = 0.007$. These analyses condition on observed scores and model panels, leaving external score uncertainty, decoding variation, and model-family dependence outside their uncertainty estimates.

The AA Intelligence Index v4.3 (Artificial Analysis, 2026) provides a general-capability reference for the eleven main-table models, excluding the assisted expert. Table 13 reports its association with each metric, where the forecasting and Capture columns track the index and the belief columns do not. Table 12 separately compares SWFI and AA against the same forecasting outcomes on matched models.

G.4 Evidence sensitivity and construction checks

Question targets are computed from recorded engine states and trajectories. Automated validation covers 1,900 decision points and 22,763 prompts, including 4,264 intervention-event prompts excluded from the reported task inventory. No violations were detected in the tested constraints. The checks cover hidden-key and future-turn leakage, observer scope, event-cell eligibility, hidden-rule evidence gates, intervention horizon bands and arm pairing, and the task-specific presence of policy summaries. Section H.3 reports the paired evidence-sensitivity ablations.

Table 13: Spearman rank correlations between the external capability index and the unrounded main-table scores across 11 unassisted model lines in each setting. All seven metrics are included; tied values receive average ranks. The assisted EXPERT line is excluded. These are descriptive model-level associations, not matched-anchor comparisons or significance tests.

Metric	OMNI-ID	EGO	OMNI-OOD
QUANTITY	+0.84	+0.67	+0.92
EVENT	+0.75	+0.86	+0.77
TRAIT	+0.07	−0.03	+0.11
RULES	+0.15	+0.31	+0.23
Capture	+0.76	+0.75	+0.58
Direction	+0.34	+0.39	+0.34
Specificity	+0.15	+0.55	+0.65

G.5 Noise sources and reload checks

Uncertainty arises from future engine randomness, ambiguity in partially observed state, reload and temporal-resolution effects, and finite evaluation samples. Repeated futures address the first source, privileged observers characterize the second under their stated information sets, and replay checks address the third. World bootstraps estimate sampling uncertainty across worlds conditional on the recorded model outputs.

All intervention truths use rerun factual arms from the same saved checkpoint rather than the uninterrupted mainline. Saves are taken every turn so short-lived events are not missed. The RNG-sharing diagnostic compares 141,250 placebo cells with 11,091 re-seeded cells: mean normalized absolute divergence is 0.116 versus 0.259, and 56.1% of placebo cells have zero divergence. Only 3,752 of 28,968 compared saved turns, or 13.0%, have identical digests. The low digest agreement shows that edited and unedited runs can follow different random streams. The divergence means describe different cell populations, so the comparison is descriptive. Placebo rollouts characterize variation introduced along the edit and reload path. Intervention errors therefore reflect both effect-estimation error and stochastic branch divergence.

H Full results

H.1 Complete three-setting results

Table 14 reports all seven metrics in OMNI-ID, EGO, and OMNI-OOD, using the score sources and references of Table 3. SWFI averages the three settings. The expert uses the shared 190-anchor sample and the knowledge and harness specified in Section G.2.4, with the same failure penalties as other models.

Table 14: Complete results across three settings. Bold denotes the best model score, excluding the expert reference. Percentages report reference-normalized completion. Missing or unparseable answers receive the failure penalties in Section F. Section G.2 specifies the numerical reference assumptions.

		L1 association		L2 belief		L3 intervention		
Line	AA Index	QUANTITY ↑	EVENT ↑	TRAIT ↑	RULES ↑	CAPTURE↑	DIRECTION↑	SPECIFICITY↑
OMNI-ID								
FLOOR	—	+0.00	+0.000	0.00	0.00	−1.00	0.00	+0.00
gpt-oss-120b	12	+0.03 (4%)	−0.031 (-29%)	0.01 (2%)	0.04 (5%)	−0.65 (25%)	0.15 (16%)	+0.05 (5%)
Gemma 4 31B	15	+0.21 (31%)	+0.012 (11%)	0.08 (12%)	0.11 (12%)	−0.61 (28%)	0.30 (33%)	**+0.19** (19%)
Nemotron 3 Ultra	23	+0.08 (12%)	+0.003 (3%)	0.08 (11%)	0.12 (13%)	−0.63 (27%)	−0.16 (-17%)	−0.33 (-33%)
Inkling	26	+0.22 (32%)	+0.012 (11%)	0.13 (20%)	0.11 (12%)	−0.31 (49%)	0.09 (10%)	−0.11 (-11%)
Qwen3.8-27B	34	+0.25 (36%)	+0.007 (7%)	0.10 (16%)	−0.11 (-12%)	−0.34 (48%)	0.24 (26%)	−0.03 (-3%)
GPT-5.6 Sol	39	+0.27 (39%)	+0.022 (20%)	**0.23** (34%)	0.16 (17%)	−0.60 (29%)	0.28 (30%)	−0.02 (-2%)
DeepSeek-V4.1-Flash	40	+0.19 (27%)	+0.017 (16%)	0.14 (22%)	**0.17** (18%)	−0.58 (31%)	0.30 (32%)	+0.07 (7%)
Qwen3.8-Flash-Next	40	+0.26 (38%)	+0.011 (10%)	0.10 (15%)	0.08 (9%)	−0.57 (31%)	0.22 (24%)	+0.05 (5%)
Kimi K3	44	+0.34 (50%)	**+0.025** (23%)	−0.00 (-0%)	0.05 (6%)	−0.35 (47%)	0.23 (25%)	+0.08 (8%)
GLM-5.3	45	**+0.37** (53%)	+0.023 (21%)	−0.01 (-2%)	0.07 (8%)	−0.25 (54%)	0.16 (17%)	−0.05 (-5%)
Claude Opus 5	45	+0.33 (48%)	+0.021 (19%)	0.17 (26%)	0.13 (14%)	**−0.08** (66%)	**0.33** (35%)	+0.12 (12%)
EXPERT	—	+0.37 (55%)	+0.021 (20%)	0.26 (39%)	0.52 (56%)	+0.13 (81%)	0.58 (63%)	+0.86 (86%)
CEILING	—	+0.68	+0.108	0.66	0.93	+0.39	0.92	1.00
EGO								
FLOOR	—	+0.00	+0.000	0.00	0.00	−1.00	0.00	+0.00
gpt-oss-120b	12	−0.20 (-29%)	−0.035 (-32%)	−0.02 (-4%)	0.06 (7%)	−0.73 (20%)	0.14 (15%)	−0.82 (-82%)
Gemma 4 31B	15	+0.05 (7%)	+0.003 (3%)	0.06 (9%)	0.11 (12%)	−0.51 (35%)	0.28 (30%)	+0.08 (8%)
Nemotron 3 Ultra	23	−0.16 (-24%)	+0.006 (6%)	−0.02 (-3%)	0.11 (12%)	−0.68 (23%)	−0.17 (-19%)	−0.39 (-39%)
Inkling	26	+0.07 (10%)	+0.008 (8%)	0.04 (6%)	0.13 (16%)	−0.31 (50%)	0.12 (14%)	−0.03 (-3%)
Qwen3.8-27B	34	+0.06 (9%)	+0.006 (5%)	0.04 (5%)	−0.20 (-23%)	−0.33 (48%)	0.24 (26%)	+0.01 (1%)
GPT-5.6 Sol	39	+0.16 (23%)	+0.019 (17%)	**0.08** (12%)	**0.15** (18%)	−0.27 (53%)	0.32 (35%)	**+0.16** (16%)
DeepSeek-V4.1-Flash	40	+0.04 (6%)	+0.014 (13%)	0.03 (5%)	0.12 (14%)	−0.69 (23%)	0.22 (24%)	+0.12 (12%)
Qwen3.8-Flash-Next	40	+0.05 (7%)	+0.009 (9%)	0.02 (3%)	0.13 (15%)	−0.39 (44%)	0.26 (28%)	+0.09 (9%)
Kimi K3	44	**+0.22** (32%)	+0.018 (17%)	0.01 (2%)	0.09 (11%)	−0.24 (54%)	0.24 (26%)	+0.03 (3%)
GLM-5.3	45	+0.12 (18%)	+0.016 (15%)	−0.02 (-2%)	0.11 (12%)	−0.21 (57%)	0.22 (24%)	+0.04 (4%)
Claude Opus 5	45	+0.19 (28%)	**+0.021** (20%)	0.03 (5%)	0.14 (16%)	**+0.01** (73%)	**0.48** (52%)	+0.10 (10%)
EXPERT	—	+0.29 (42%)	+0.017 (16%)	0.14 (22%)	0.38 (44%)	+0.15 (83%)	0.64 (69%)	+0.84 (84%)
CEILING	—	+0.68	+0.106	0.66	0.86	+0.39	0.92	1.00
OMNI-OOD								
FLOOR	—	+0.00	+0.000	0.00	0.00	−1.00	0.00	+0.00
gpt-oss-120b	12	−0.01 (-1%)	−0.030 (-28%)	0.07 (10%)	0.10 (11%)	−0.69 (24%)	0.08 (9%)	−0.10 (-10%)
Gemma 4 31B	15	+0.13 (19%)	+0.008 (8%)	0.16 (24%)	0.10 (11%)	−0.61 (30%)	0.20 (22%)	+0.02 (2%)
Nemotron 3 Ultra	23	+0.05 (7%)	+0.000 (0%)	0.04 (7%)	−0.01 (-1%)	−0.80 (16%)	−0.19 (-21%)	−0.29 (-29%)
Inkling	26	+0.20 (30%)	+0.012 (11%)	0.19 (29%)	0.05 (6%)	−0.42 (45%)	−0.04 (-4%)	−0.12 (-12%)
Qwen3.8-27B	34	+0.27 (40%)	+0.002 (2%)	0.15 (23%)	−0.22 (-24%)	−0.41 (45%)	0.16 (18%)	+0.01 (1%)
GPT-5.6 Sol	39	+0.29 (43%)	+0.021 (20%)	**0.30** (45%)	0.18 (20%)	−0.24 (58%)	**0.41** (44%)	+0.20 (20%)
DeepSeek-V4.1-Flash	40	+0.21 (30%)	+0.016 (16%)	0.24 (36%)	**0.19** (21%)	−0.69 (24%)	0.19 (21%)	**+0.44** (44%)
Qwen3.8-Flash-Next	40	+0.28 (41%)	+0.007 (6%)	0.17 (25%)	−0.02 (-3%)	−0.63 (29%)	0.17 (18%)	+0.04 (4%)
Kimi K3	44	+0.36 (53%)	**+0.021** (20%)	0.06 (9%)	0.14 (15%)	−0.45 (42%)	0.23 (25%)	+0.04 (4%)
GLM-5.3	45	**+0.41** (60%)	+0.021 (20%)	−0.04 (-7%)	0.04 (4%)	−0.33 (51%)	0.11 (12%)	+0.02 (2%)
Claude Opus 5	45	+0.33 (48%)	+0.017 (17%)	0.26 (40%)	0.18 (20%)	**−0.10** (70%)	0.19 (21%)	+0.08 (8%)
EXPERT	—	+0.37 (54%)	+0.023 (22%)	0.29 (45%)	0.55 (60%)	+0.06 (81%)	0.34 (37%)	+0.85 (85%)
CEILING	—	+0.68	+0.105	0.66	0.90	+0.30	0.92	1.00

H.2 Detailed model profiles

Figure 8 compares the six quantity families across all models and settings. Overall-score increments remain difficult despite positive skill on many stock quantities. The diagnostic equalizes variables within families as specified in Section F. Tables 15 and 16 further show that hidden-target recovery varies across attributes. Table 14 shows model-dependent changes under OOD shift: Flash-Next improves in quantity forecasting, Opus 5 changes little, and Gemma 4 declines. The OOD set changes rulesets and trajectory budgets without matching ID difficulty or selecting worlds to enforce a particular score ordering.

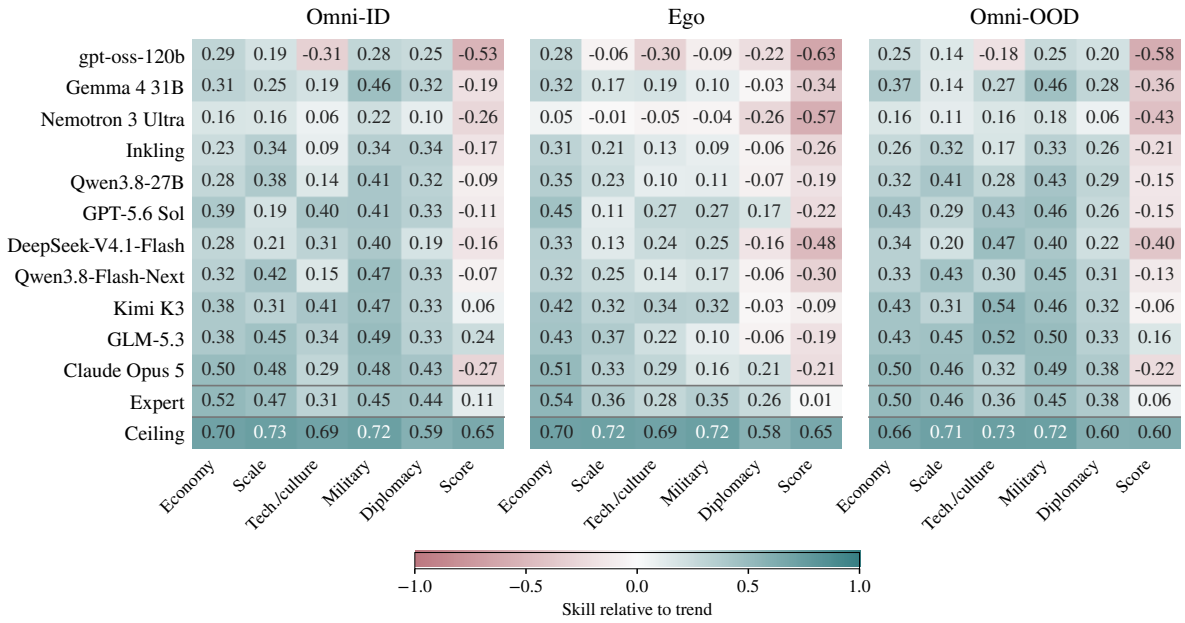


Figure 8: Quantity-family skill across three settings. All panels share a color scale centered on the trend reference at zero. Variables are equally weighted within each family.

Table 15: Pairwise ranking accuracy by personality trait. Triples are Omni-ID / Ego / Omni-OOD, using the main-table samples and failure penalties.

Model	Aggressive	Expansionist	Trader	Builder
gpt-oss-120b	0.51 / 0.49 / 0.52	0.51 / 0.51 / 0.56	0.49 / 0.49 / 0.49	0.50 / 0.49 / 0.49
Gemma 4 31B	0.52 / 0.51 / 0.54	0.53 / 0.53 / 0.57	0.52 / 0.51 / 0.52	0.52 / 0.51 / 0.52
Nemotron 3 Ultra	0.51 / 0.48 / 0.50	0.52 / 0.52 / 0.55	0.53 / 0.50 / 0.52	0.51 / 0.49 / 0.48
Inkling	0.53 / 0.50 / 0.55	0.54 / 0.53 / 0.58	0.54 / 0.50 / 0.54	0.52 / 0.50 / 0.52
Qwen3.8-27B	0.52 / 0.50 / 0.53	0.53 / 0.53 / 0.57	0.52 / 0.50 / 0.52	0.53 / 0.51 / 0.52
GPT-5.6 Sol	0.54 / 0.52 / 0.54	0.55 / 0.49 / 0.61	0.59 / 0.54 / 0.58	0.55 / 0.53 / 0.56
DeepSeek-V4.1-Flash	0.51 / 0.49 / 0.54	0.54 / 0.53 / 0.59	0.56 / 0.51 / 0.56	0.54 / 0.50 / 0.54
Qwen3.8-Flash-Next	0.51 / 0.49 / 0.53	0.53 / 0.53 / 0.57	0.53 / 0.49 / 0.53	0.52 / 0.51 / 0.53
Kimi K3	0.48 / 0.49 / 0.50	0.50 / 0.52 / 0.55	0.52 / 0.50 / 0.52	0.50 / 0.50 / 0.49
GLM-5.3	0.47 / 0.48 / 0.47	0.51 / 0.51 / 0.52	0.52 / 0.50 / 0.50	0.49 / 0.50 / 0.47
Claude Opus 5	0.54 / 0.50 / 0.55	0.53 / 0.51 / 0.60	0.57 / 0.54 / 0.56	0.53 / 0.48 / 0.54
EXPERT	0.57 / 0.55 / 0.56	0.58 / 0.53 / 0.62	0.57 / 0.53 / 0.54	0.53 / 0.52 / 0.56

Table 16: Classification accuracy by world rule. Triples are Omni-ID / Ego / Omni-OOD, using the main-table samples and failure penalties.

Model	Disaster frequency	Growth cost	Research cost
gpt-oss-120b	0.63 / 0.60 / 0.61	0.34 / 0.35 / 0.37	0.32 / 0.36 / 0.41
Gemma 4 31B	0.65 / 0.64 / 0.62	0.39 / 0.38 / 0.34	0.36 / 0.37 / 0.41
Nemotron 3 Ultra	0.63 / 0.62 / 0.62	0.39 / 0.36 / 0.30	0.40 / 0.41 / 0.28
Inkling	0.65 / 0.64 / 0.62	0.36 / 0.39 / 0.35	0.39 / 0.40 / 0.34
Qwen3.8-27B	0.42 / 0.38 / 0.40	0.29 / 0.25 / 0.24	0.31 / 0.27 / 0.21
GPT-5.6 Sol	0.73 / 0.66 / 0.66	0.32 / 0.37 / 0.44	0.43 / 0.43 / 0.42
DeepSeek-V4.1-Flash	0.68 / 0.65 / 0.67	0.40 / 0.38 / 0.40	0.42 / 0.38 / 0.46
Qwen3.8-Flash-Next	0.56 / 0.61 / 0.55	0.41 / 0.40 / 0.32	0.39 / 0.41 / 0.31
Kimi K3	0.62 / 0.63 / 0.63	0.35 / 0.37 / 0.39	0.33 / 0.36 / 0.42
GLM-5.3	0.58 / 0.61 / 0.54	0.37 / 0.36 / 0.34	0.38 / 0.41 / 0.40
Claude Opus 5	0.71 / 0.64 / 0.63	0.37 / 0.33 / 0.39	0.36 / 0.46 / 0.50
EXPERT	0.77 / 0.76 / 0.72	0.66 / 0.46 / 0.77	0.68 / 0.61 / 0.63

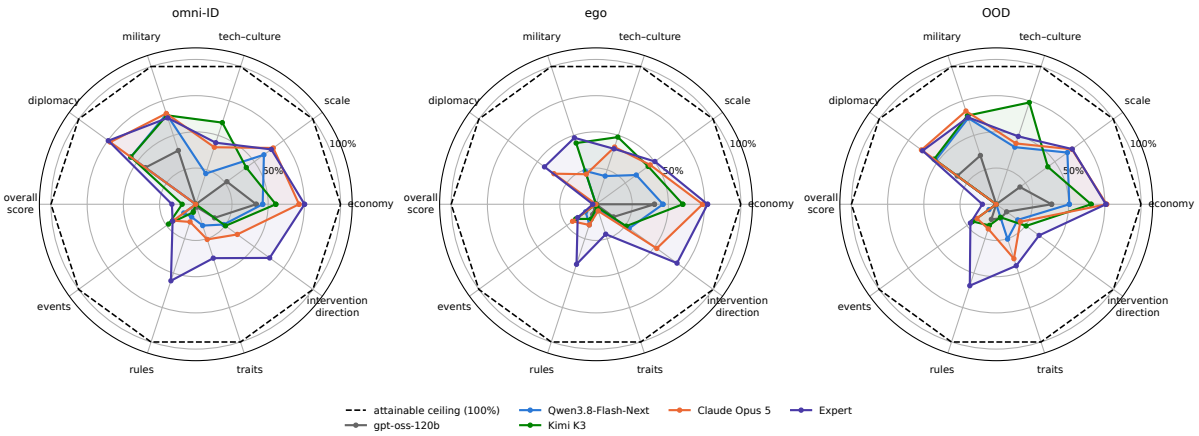


Figure 9: Selected model profiles across three settings. Axes show reference-normalized scores clipped to $[0, 1]$ for display. The intervention axis uses direction, and radar area is not an aggregate score.

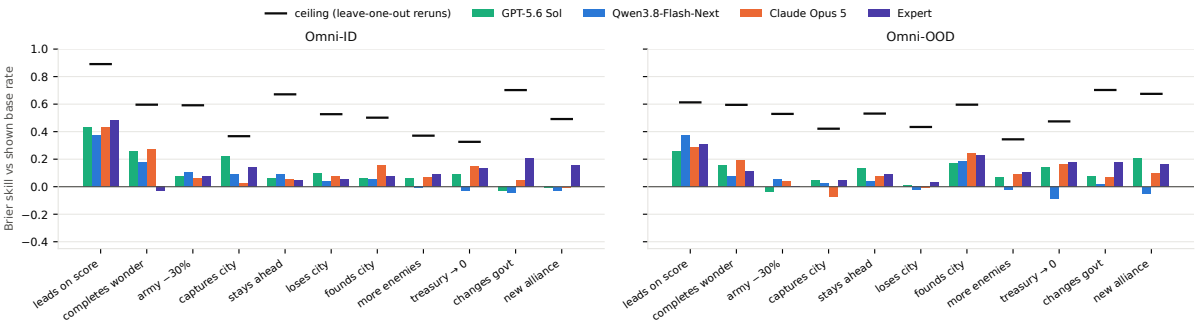


Figure 10: Event-template Brier skill ratios in OMNI-ID and OMNI-OOD, with leave-one-out rerun references in black. This diagnostic uses $1 - \sum \text{Brier} / \sum \text{Brier}_{\text{ref}}$, not the headline Brier gain.

H.3 Information ablations and three-game study notes

Paired design. The Flash-Next ablations use 60 basic worlds, or 240 anchors, in OMNI-ID, except the primer, which covers 40 worlds. Each row of Table 17 uses its own common valid cells and paired world-bootstrap interval, so baselines can differ across rows. The conditions remove history or thinking, add generic rules, supply privileged future narratives, or add study notes. Future narratives provide an information control rather than a deployable forecast input.

Study input and notes. The one-shot study uses three independently seeded, 3-civilization, 400-turn games outside the evaluation corpus. Each is summarized at turns 60, 145, 230, 315, and 400, with intervening narratives. The 411,688-character input produces an 11,221-character note reused across 1,624 evaluation prompts. No model parameters are updated, and the note is not revised during evaluation.

Forecasting gains. Table 17 shows a paired quantity gain of +0.086 with interval [+0.061, +0.111]. Notes also improve event Brier gain from -0.034 to -0.015. This event comparison uses the no-base-rate protocol, including eligible leader items. Supplying base rates is a separate condition that reaches $+0.012$ from the same baseline. Table 18 shows similar gains for real and placebo capture, both remaining below zero. Notes also change the no-intervention forecast. Thus, learned context improves forecasting broadly, while selective responses to intervention mechanisms remain difficult.

H.4 Intervention breakdowns and mechanism cards

Direction-conditioned magnitude distributions. Figure 4 examines effect magnitudes among correct-sign predictions under the main score's real-arm, target-variable, horizon-band, and truth-movement rules. Predicted and realized effects use the transformed, normalized differences in Section 4.1. Eligible cells exclude undefined control-relative loss ratios, including zero-loss control forecasts, with counts in Table 22. Diagnostic direction accuracy is the cell-weighted fraction of correct signs among valid nonzero responses. The main score additionally penalizes missing answers and weights operator scores by eligible-cell counts. Tables 19 to 21 report missing and zero responses separately. Relative magnitude is $|\widehat{\delta}|/|\delta|$, where $\widehat{\delta}$ and δ denote predicted and realized effects. Figure medians and 25th–75th percentile bars summarize its distribution across correct-sign cells, with 100% denoting a magnitude match. Ratios are cell-weighted and unclipped, with linearly interpolated quantiles. Additional bins classify predictions below half, within a factor of two, or above twice the realized magnitude, using inequalities in the same scoring space. The conditional score S_+ applies the clipped equal-operator Capture formula to correct-sign cells, retaining operators with at least three such cells.

Conditional results. Table 23 shows that near-matching medians coexist with broad magnitude distributions across settings. In OMNI-ID, six of eleven models have negative S_+ estimates, with intervals entirely below zero for gpt-oss, Gemma, and DeepSeek. For these models, magnitude errors persist even when the direction is correct. GLM-5.3 has positive S_+ with an interval above zero, while its main Capture remains negative and its direction intervals exceed chance in all three settings. Its conditional improvement illustrates the contribution of wrong-direction predictions to overall error. A correct-sign prediction above twice the realized magnitude has greater absolute error than a zero-effect prediction. Table 22 shows that these overshoots persist, at lower frequencies, after removing effects at or below each operator's lower quartile before conditioning on correct signs. Underestimation and overestimation therefore remain relevant beyond the smallest effects, motivating state-conditioned magnitude calibration.

Sampling and interpretation. The magnitude bars describe dispersion across predictions, and intervals follow Section F. Model-specific samples and correct-sign subsets differ, so these comparisons diagnose within-model errors. Conditioning also changes the cells and operators included in S_+ relative to main Capture. The targets are realized stochastic branch differences, and future calibration studies should

Table 17: Paired information ablations of Flash-Next in Omni-ID. Each row uses its own common valid cells; intervals are paired world bootstraps. Event comparisons here use a no-base-rate baseline.

Condition	Task	Worlds	Cells	Base	Added/ablated	Change	95% interval
One frame	QUANTITY	60	27251	+0.269	+0.137	-0.132	[-0.163, -0.103]
Future narrative	QUANTITY	60	23085	+0.285	+0.346	+0.061	[+0.031, +0.091]
Rules primer	QUANTITY	40	18027	+0.266	+0.247	-0.019	[-0.050, +0.011]
Study notes	QUANTITY	60	26408	+0.268	+0.354	+0.086	[+0.061, +0.111]
Thinking off	QUANTITY	60	27564	+0.270	+0.217	-0.053	[-0.078, -0.029]
Study notes	EVENT	60	22018	-0.034	-0.015	+0.019	[+0.014, +0.025]
Base rate supplied	EVENT	60	22018	-0.034	+0.012	+0.046	[+0.041, +0.052]

Table 18: Study notes: paired CAPTURE on each arm's common valid moved cells in Omni-ID. Both arm and no-intervention predictions are note-conditioned. Intervals use paired world Bayesian bootstrap weights.

Arm	Cells	Worlds	Ops	Base	Notes	Change	95% interval
real	433	60	11	-0.514	-0.277	+0.237	[+0.070, +0.343]
placebo	341	58	11	-0.585	-0.335	+0.250	[+0.053, +0.357]

evaluate gains on held-out worlds.

Mechanism-card design. The mechanism-card study contains 60 selected anchors and 310 intervention prompts, with a note on 260 real/placebo rows across 11 operators. Each note states generic engine mechanics, is identical across worlds for its operator, and is inserted immediately before the task block. The same note is used for real and placebo edits. No future arm outcome is supplied. For the paired analysis, both conditions use the corresponding model's main-experiment no-intervention median. Holding this reference fixed attributes paired changes to the edited-arm forecasts. Figure 11 reports these common-control comparisons.

Mechanism-card results. Figure 11 shows that Qwen3.8-27B improves in OMNI-ID, while Flash-Next and Gemma decline. All three remain below the zero-effect reference. Flash-Next and Qwen3.8-27B improve in EGO, whereas OOD gains do not persist. Responses to mechanism information thus depend on both the model and evaluation setting. In OMNI-ID, common valid real-arm cells number 48, 48, and 52 for Flash-Next, Qwen3.8-27B, and Gemma, covering 7, 7, and 8 operators, respectively. Pairing is performed within each model. Only Flash-Next's OMNI-ID real-arm comparison has at least two worlds per operator, giving a 95% paired world-bootstrap interval of $[-0.170, +0.028]$. Each of the other 17 comparisons includes a singleton-world operator. Table 25 reports their point estimates, with between-world intervals omitted under the same sampling rule.

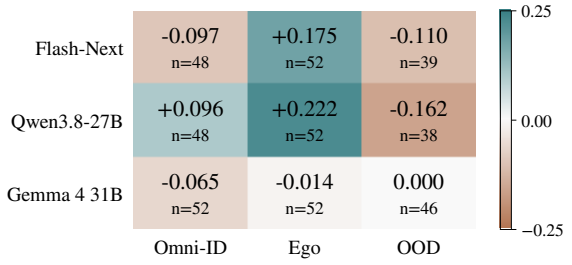


Figure 11: Mechanism-card effects with fixed control forecasts. Colors show Capture changes and counts give matched cells.

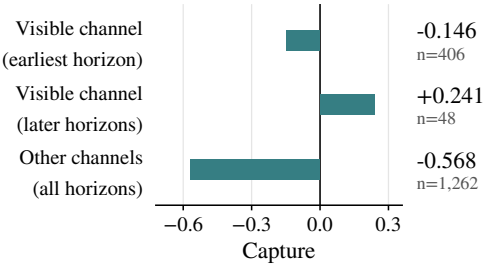


Figure 12: Flash-Next's OMNI-ID Capture by channel and horizon group. Operator compositions differ across the three groups.

Table 19: Direction-conditioned magnitude diagnostics in omni-ID. Counts give eligible, missing, zero-response, and correct-sign cells. Direction is the percentage correct among valid nonzero responses. Magnitude-bin percentages condition on a correct sign. Brackets give 95% world-bootstrap intervals. S_+ is selected-subset capture, not a replacement main score. An asterisk marks an omitted interval due to a singleton-world operator.

Model	Counts	Direction	$< 0.5\times$	0.5–2×	$> 2\times$	S_+
gpt-oss-120b	1749/57/105/937	59 [56, 62]	21	45	34 [30, 38]	-0.34 [-0.44, -0.16]
Gemma 4 31B	1759/1/145/1033	64 [61, 67]	18	48	34 [31, 38]	-0.31 [-0.41, -0.20]
Nemotron 3 Ultra	1767/545/93/717	64 [60, 67]	15	52	33 [29, 37]	-0.07 [-0.20, +0.03]
Inkling	1772/169/178/871	61 [58, 65]	27	51	21 [18, 25]	+0.20 [+0.10, +0.28]
Qwen3.8-27B	1764/56/100/1017	63 [60, 66]	23	50	27 [24, 30]	+0.03 [-0.07, +0.13]
GPT-5.6 Sol	176/0/16/103	64 [54, 74]	9	57	34 [25, 44]	-0.07 [-0.30, +0.18]
DeepSeek-V4.1-Flash	1769/4/164/1027	64 [61, 67]	17	48	35 [32, 38]	-0.27 [-0.38, -0.15]
Flash-Next	1768/52/77/1018	62 [59, 65]	26	47	27 [24, 31]	-0.11 [-0.18, +0.02]
Kimi K3	1774/8/123/1006	61 [58, 64]	19	53	28 [25, 32]	+0.08 [-0.04, +0.17]
GLM-5.3	1773/142/142/937	63 [60, 66]	26	52	22 [19, 25]	+0.16 [+0.10, +0.26]
Claude Opus 5	176/4/35/94	69 [58, 78]	33	50	17 [10, 26]	+0.32 [+0.06, +0.46]
Expert	176/0/59/90	77 [65, 87]	36	57	8 [3, 14]	+0.33 [+0.23, +0.41]

Table 20: Direction-conditioned magnitude diagnostics in ego. Counts give eligible, missing, zero-response, and correct-sign cells. Direction is the percentage correct among valid nonzero responses. Magnitude-bin percentages condition on a correct sign. Brackets give 95% world-bootstrap intervals. S_+ is selected-subset capture, not a replacement main score. An asterisk marks an omitted interval due to a singleton-world operator.

Model	Counts	Direction	$< 0.5\times$	0.5–2×	$> 2\times$	S_+
gpt-oss-120b	1753/25/132/917	57 [54, 61]	20	48	32 [28, 36]	-0.33 [-0.44, -0.20]
Gemma 4 31B	1767/2/185/995	63 [60, 66]	24	48	27 [24, 30]	-0.10 [-0.20, +0.02]
Nemotron 3 Ultra	1767/480/104/700	59 [55, 63]	16	47	37 [33, 41]	-0.24 [-0.36, -0.13]
Inkling	1774/135/218/885	62 [59, 66]	27	49	24 [20, 27]	+0.12 [+0.04, +0.23]
Qwen3.8-27B	1770/11/116/1014	62 [59, 65]	22	50	29 [25, 32]	+0.04 [-0.07, +0.16]
GPT-5.6 Sol	176/0/14/107	66 [56, 75]	17	60	23 [15, 35]	+0.11 [-0.10, +0.28]
DeepSeek-V4.1-Flash	1772/20/157/977	61 [58, 64]	15	48	37 [33, 40]	-0.32 [-0.43, -0.18]
Flash-Next	1774/13/119/1028	63 [60, 66]	25	48	27 [23, 30]	-0.09 [-0.16, +0.03]
Kimi K3	1774/4/123/1007	61 [58, 64]	22	53	26 [22, 29]	+0.14 [+0.06, +0.20]
GLM-5.3	1774/60/130/992	63 [60, 66]	25	54	21 [18, 24]	+0.18 [+0.11, +0.26]
Claude Opus 5	176/0/28/108	73 [63, 81]	30	58	12 [6, 19]	+0.37 [+0.26, +0.45]
Expert	176/0/51/97	78 [66, 87]	35	56	9 [4, 15]	+0.32 [+0.21, +0.38]

Channel and horizon diagnostic. Figure 12 keeps the formal target variables, horizon bands, and moved-cell mask. Its shortest horizon is determined from asked target cells before filtering for truth movement or valid model output. Visible-channel variables are treasury for the gold edit, technologies learned for research points or a technology grant, and city count/total city size for city destruction, subject to each operator's formal scored-variable mask. We partition the eligible cells into three disjoint groups: the designated visible channels at the earliest eligible horizon, covering 406 cells from three operators, the same channels at later eligible horizons, covering 48 cells from city destruction only, and all other channels at any eligible horizon, covering 1,262 cells from nine operators. Earliest/later is relative to each world, anchor, and operator, not a global short/long cutoff. If the earliest prediction is missing or its truth is unmoved, a later prediction does not become an earliest-horizon cell. The technology-grant operator has no eligible moved cell in either visible group and receives no score there. Other channels include remaining variables under visible-state edits as well as responses to hidden-state edits. All three groups contain observable quantity forecasts in the global-view setting.

The positive later-horizon score concerns city destruction alone. Differences across these groups therefore combine horizon and operator composition. The direct-effect sign check in Section O complements this analysis with a broader target set, including city count outside the main Capture mask.

Table 21: Direction-conditioned magnitude diagnostics in OOD. Counts give eligible, missing, zero-response, and correct-sign cells. Direction is the percentage correct among valid nonzero responses. Magnitude-bin percentages condition on a correct sign. Brackets give 95% world-bootstrap intervals. S_+ is selected-subset capture, not a replacement main score. An asterisk marks an omitted interval due to a singleton-world operator.

Model	Counts	Direction	$< 0.5\times$	0.5–2×	$> 2\times$	S_+
gpt-oss-120b	1781/118/111/903	58 [55, 61]	16	42	42 [38, 46]	-0.42 [-0.52, -0.28]
Gemma 4 31B	1793/2/163/971	60 [57, 62]	16	40	44 [41, 48]	-0.31 [-0.40, -0.24]
Nemotron 3 Ultra	1802/592/66/716	63 [59, 67]	15	41	44 [40, 49]	-0.29 [-0.44, -0.15]
Inkling	1804/228/137/812	56 [53, 59]	21	52	28 [24, 31]	+0.04 [-0.09, +0.16]
Qwen3.8-27B	1802/80/87/991	61 [58, 64]	16	46	39 [35, 42]	-0.13 [-0.22, -0.00]
GPT-5.6 Sol	169/0/12/108	69 [59, 79]	10	46	44 [33, 56]	-0.07 [-0.27, +0.11]
DeepSeek-V4.1-Flash	1803/25/167/967	60 [57, 63]	14	43	43 [39, 47]	-0.36 [-0.48, -0.23]
Flash-Next	1802/69/84/998	61 [58, 64]	19	43	38 [35, 42]	-0.21 [-0.33, -0.05]
Kimi K3	1802/12/114/1030	61 [59, 65]	16	46	37 [34, 41]	-0.06 [-0.15, +0.03]
GLM-5.3	1807/101/134/929	59 [56, 62]	22	50	28 [25, 31]	+0.08 [-0.01, +0.20]
Claude Opus 5	169/0/18/90	60 [49, 70]	27	50	23 [13, 36]	+0.25 [+0.11, +0.37]
Expert	169/0/59/73	66 [53, 79]	48	37	15 [8, 27]	+0.23 [+0.13, +0.33]

Table 22: Magnitude sensitivity in OMNI-ID. All percentages condition on a correct sign. The sensitivity filter removes effects at or below each operator's lower quartile before selecting correct-sign predictions. Excluded counts refer instead to valid predictions omitted by the inherited main scoring mask.

Model	$< 1\times$	$> 1\times$	$> 2\times$	Filtered $> 2\times$	Filtered n_+	Excluded
gpt-oss-120b	38.1	52.7	33.7	27.6	646	25
Gemma 4 31B	38.7	54.1	34.1	25.6	738	15
Nemotron 3 Ultra	33.5	57.3	33.3	25.7	501	7
Inkling	50.9	40.4	21.4	12.5	615	2
Qwen3.8-27B	42.1	49.7	26.8	17.5	718	10
GPT-5.6 Sol	25.2	68.0	34.0	23.5	68	0
DeepSeek-V4.1-Flash	35.6	57.9	35.0	27.1	730	5
Flash-Next	44.5	47.6	27.2	18.6	714	6
Kimi K3	40.5	50.6	28.3	19.9	695	0
GLM-5.3	47.5	43.3	22.1	13.6	664	1
Claude Opus 5	52.1	39.4	17.0	7.7	65	0
Expert	58.9	36.7	7.8	3.1	65	0

Table 23: Relative effect magnitudes conditional on correct direction. Entries give the median and [25th, 75th] percentiles, all in percentages. The ratio uses absolute predicted and realized effects in the scoring space. 100% denotes a magnitude match. Brackets describe sample dispersion, not confidence intervals.

Model	OMNI-ID	EGO	OMNI-OOD
gpt-oss-120b	120.0 [58.3, 300.0]	106.2 [55.7, 300.0]	158.0 [68.4, 455.4]
Gemma 4 31B	122.2 [62.5, 300.0]	100.0 [50.0, 229.7]	172.8 [82.1, 566.6]
Nemotron 3 Ultra	123.3 [73.8, 300.0]	140.0 [68.6, 327.5]	183.3 [83.3, 407.8]
Inkling	96.4 [43.4, 180.0]	100.0 [45.1, 200.0]	105.3 [53.1, 246.1]
Qwen3.8-27B	100.0 [50.0, 225.0]	108.5 [52.1, 244.0]	150.0 [82.0, 349.0]
GPT-5.6 Sol	160.0 [99.7, 272.1]	133.3 [69.8, 200.0]	176.0 [99.0, 406.6]
DeepSeek-V4.1-Flash	130.8 [66.7, 300.0]	136.7 [72.2, 333.3]	162.5 [81.8, 468.3]
Flash-Next	100.0 [46.9, 233.3]	100.0 [50.0, 233.3]	140.0 [63.8, 328.4]
Kimi K3	103.9 [57.8, 250.0]	100.0 [50.5, 210.6]	140.8 [66.7, 329.0]
GLM-5.3	100.0 [45.5, 200.0]	100.0 [47.1, 175.9]	100.0 [54.5, 235.3]
Claude Opus 5	85.7 [33.3, 161.4]	100.0 [40.0, 150.0]	100.0 [40.2, 200.0]
Expert	75.7 [22.2, 157.5]	74.3 [23.5, 157.5]	50.0 [18.2, 130.8]

Table 24: Intervention diagnostics across three settings. Capture and shift are reported for real arms with matched placebos in parentheses. Shift is the mean absolute change from the model's no-intervention forecast in the normalized units of Section 4.1. Specificity applies the missing-response penalty to the clipped shift ratio defined in Equation (4) and Section F. Triples are OMNI-ID / EGO / OMNI-OOD.

Line	capture real (placebo)	direction	shift real (placebo)
gpt-oss-120b	−0.65 (−0.74) / −0.73 (−0.77) / −0.69 (−0.87)	0.57 / 0.56 / 0.54	0.56 (0.50) / 0.63 (1.15) / 0.58 (0.60)
Gemma 4 31B	−0.61 (−0.62) / −0.51 (−0.56) / −0.61 (−0.69)	0.64 / 0.63 / 0.59	0.38 (0.30) / 0.29 (0.27) / 0.41 (0.40)
Nemotron 3 Ultra	−0.63 (−0.74) / −0.68 (−0.83) / −0.80 (−0.87)	0.43 / 0.42 / 0.41	0.28 (0.28) / 0.31 (0.29) / 0.28 (0.24)
Inkling	−0.31 (−0.43) / −0.31 (−0.44) / −0.42 (−0.55)	0.54 / 0.56 / 0.48	0.15 (0.15) / 0.18 (0.17) / 0.14 (0.14)
Qwen3.8-27B	−0.34 (−0.50) / −0.33 (−0.47) / −0.41 (−0.66)	0.61 / 0.61 / 0.57	0.20 (0.20) / 0.27 (0.26) / 0.18 (0.17)
GPT-5.6 Sol	−0.60 (−0.70) / −0.27 (−0.60) / −0.24 (−0.50)	0.63 / 0.65 / 0.69	0.17 (0.17) / 0.17 (0.15) / 0.15 (0.12)
DeepSeek-V4.1-Flash	−0.58 (−0.66) / −0.69 (−0.69) / −0.69 (−0.86)	0.64 / 0.60 / 0.59	0.26 (0.24) / 0.31 (0.27) / 0.39 (0.21)
Qwen3.8-Flash-Next	−0.57 (−0.56) / −0.39 (−0.57) / −0.63 (−0.74)	0.60 / 0.62 / 0.58	0.25 (0.23) / 0.31 (0.28) / 0.22 (0.21)
Kimi K3	−0.35 (−0.52) / −0.24 (−0.50) / −0.45 (−0.68)	0.61 / 0.61 / 0.61	0.18 (0.16) / 0.18 (0.17) / 0.14 (0.14)
GLM-5.3	−0.25 (−0.47) / −0.21 (−0.39) / −0.33 (−0.52)	0.57 / 0.60 / 0.55	0.13 (0.12) / 0.16 (0.15) / 0.11 (0.10)
Claude Opus 5	−0.08 (−0.26) / +0.01 (−0.04) / −0.10 (−0.00)	0.65 / 0.72 / 0.59	0.07 (0.06) / 0.07 (0.07) / 0.06 (0.05)
EXPERT	+0.13 (+0.00) / +0.15 (+0.01) / +0.06 (+0.01)	0.76 / 0.79 / 0.65	0.00 (0.00) / 0.00 (0.00) / 0.00 (0.00)

Table 25: Paired Capture with mechanism cards and a shared no-intervention prediction. Counts denote cells/operators. Real and placebo arms use separate moved-cell sets.

Model	Setting	Arm	Cells/ops	Base	Card	Change
Flash-Next	omni-ID	real	48/7	-0.581	-0.678	-0.097
Flash-Next	omni-ID	placebo	44/8	-0.436	-0.626	-0.190
Flash-Next	ego	real	52/8	-0.406	-0.231	+0.175
Flash-Next	ego	placebo	44/8	-0.349	-0.237	+0.112
Flash-Next	OOD	real	39/9	-0.482	-0.592	-0.110
Flash-Next	OOD	placebo	15/4	-0.251	-0.372	-0.121
Qwen3.8-27B	omni-ID	real	48/7	-0.327	-0.231	+0.096
Qwen3.8-27B	omni-ID	placebo	44/8	-0.431	-0.458	-0.027
Qwen3.8-27B	ego	real	52/8	-0.436	-0.214	+0.222
Qwen3.8-27B	ego	placebo	44/8	-0.614	-0.435	+0.179
Qwen3.8-27B	OOD	real	38/9	-0.300	-0.462	-0.162
Qwen3.8-27B	OOD	placebo	21/4	-0.772	-0.743	+0.029
Gemma 4 31B	omni-ID	real	52/8	-0.521	-0.586	-0.065
Gemma 4 31B	omni-ID	placebo	44/8	-0.547	-0.639	-0.092
Gemma 4 31B	ego	real	52/8	-0.450	-0.463	-0.014
Gemma 4 31B	ego	placebo	44/8	-0.396	-0.408	-0.012
Gemma 4 31B	OOD	real	46/9	-0.595	-0.595	-0.000
Gemma 4 31B	OOD	placebo	24/4	-0.663	-0.800	-0.137

I Process verification protocols and diagnostics

I.1 Scope and output protocols

The aligned process study evaluates Flash-Next, Qwen3.8-27B, Gemma 4 31B, and Kimi K3 on the same 190 anchors, of which 90 are Omni-ID and 100 OOD, and GPT-5.6 Sol on 30 of these anchors, 17 Omni-ID and 13 OOD. Sol reuses compatible Free responses on 22 overlapping anchors and adds responses on eight selected by budget and phase metadata. Table and Code cover all 30. Each model contributes its full evaluated sample. Free and Table also run the egocentric condition. Code has no masked structured-input implementation. Main-text comparisons average Omni-ID and OOD equally.

Free and Table. Free produces ordinary answers with unrestricted reasoning and has no prescribed process score. Table adds explicit intermediate predictions before answering. For quantitative questions, the aligned protocol prescribes cells near one-quarter, one-half, and three-quarters of each horizon, excluding all answer times. The target and quantities are selected from question metadata before inference. Table must include these cells and may add up to eight other claims. Its claim types cover quantities, diplomatic states, governments, technologies, city ownership, and wonders. Other questions retain the general 8–40-line requirement. Numeric process scores use the prescribed intermediate cells.

Code and feedback. Code receives structured observation frames and the ruleset name, with the intervention applied when appropriate. It writes `predict(world, rng)`, whose 50 executions yield answer quantiles, event frequencies, and intermediate predictions. Up to four generations allow execution repairs and back-testing from an earlier observed frame to the last observed frame. Future evaluation truth is never returned as feedback. Back-testing stops when at most the larger of two cells or 15% of checked cells disagree, or after two rounds without improvement. The retained runnable round minimizes disagreement. The comparison evaluates complete reasoning protocols, including execution and feedback.

I.2 Scoring units and aggregation

A state unit specifies an anchor, question, civilization, quantity, and answer horizon. Intermediate errors average transformed, scale-normalized absolute errors at the prescribed times. Table supplies point predictions, while Code supplies the mean of its sampled intermediate states. Final point errors use the answer median. Table and Code are paired on common valid units within each model and scope. L1 uses ordinary quantity questions. L3-state uses real-intervention branch states. This paired diagnostic includes only units with available truth and valid predictions in both protocols.

Answer-positive (A+) means final point error is strictly below its trend-reference error. Process-positive (P+) applies the same criterion to intermediate errors. P- denotes an intermediate error at or above the reference. The joint A+P- fraction is the number of above-reference answers with intermediate forecasts at or below reference performance, divided by all eligible units.

Intermediate-state skill is computed within each world and quantity family as $\mathrm{clip}(1 - \sum e / \sum e_{\mathrm{trend}}, -1, 1)$ for positive reference denominators, averaged over families and then worlds. Both this skill and quadrant shares are averaged equally over Omni-ID and OOD. Table and Code use the same valid units within a model, although eligible sets differ across models. These diagnostics evaluate intermediate states and final point forecasts. Formal Quantity and Capture scores separately assess forecast distributions and intervention effects.

Figure 5a additionally matches Free final answers to these Table/Code units. Within each model, all three formats use the same eligible cells, excluding missing Free answers and nonfinite errors. Bar heights give the fraction of final median errors strictly below the trend reference, averaged equally across L1 and L3 branch-state scopes and then Omni-ID and OOD. Hatched segments give the joint A+P- fraction on the

Table 26: Completed output census and diagnostic coverage. F/T/C counts include all executed settings. Missing state/effect units are counted before protocol matching.

Model	Anchors	Outputs F/T/C	State missing	Effect missing	Runnable
Qwen3.8-27B	190	1537/1537/1034	200/4272	142/2320	1034/1034
Gemma 4 31B	190	1537/1537/1034	205/4272	157/2320	1033/1034
Flash-Next	190	1537/1537/1034	205/4272	159/2320	1033/1034
Kimi K3	190	1537/1537/1034	238/4272	180/2320	1033/1034
GPT-5.6 Sol	30	269/269/169	48/728	29/400	169/169

Table 27: Intermediate-state skill and joint answer-positive/process-negative shares. Percentages use all eligible matched units. L3-state evaluates intervention-branch states.

Model	Protocol	L1 skill	L3-state skill	L1 A+P- (%)	L3-state A+P- (%)
Qwen3.8-27B	Table	-0.027	$+0.067$	10.55	5.66
Qwen3.8-27B	Code	-0.006	-0.049	5.84	4.82
Gemma 4 31B	Table	-0.058	-0.036	10.01	5.92
Gemma 4 31B	Code	-0.013	-0.050	7.38	4.49
Flash-Next	Table	-0.023	$+0.028$	7.63	9.16
Flash-Next	Code	-0.065	-0.001	6.49	4.54
Kimi K3	Table	$+0.021$	$+0.017$	10.32	8.53
Kimi K3	Code	$+0.000$	$+0.153$	5.65	5.08
GPT-5.6 Sol	Table	$+0.064$	$+0.115$	3.08	8.17
GPT-5.6 Sol	Code	$+0.035$	$+0.032$	7.35	2.08

same denominator. The combined diagnostic concerns numerical point forecasts, with event forecasting evaluated separately.

Formal final-answer scores retain the benchmark's existing scorer and reference cells. L3 Capture averages operator-level effect skills under the frozen eligibility, clipping, and control definitions. Its reported value is the equal Omni-ID/OOD mean. Raw Capture measures improvement over predicting no effect and can be negative.

I.3 Model-level process quality and Capture

Figure 5b shows a positive model-level association between L3 branch-state process skill and final Capture completion. Each score averages Omni-ID and OOD equally, then Table and Code equally, yielding one point per model, with Free excluded. Table 29 compares model-averaged, pooled, and protocol-specific associations. The L1 comparison pairs L1 process skill with equally weighted, reference-normalized Quantity and Event scores. The model-mean Pearson correlation is 0.69 and Spearman correlation is 0.80. These associations support intermediate-state prediction quality as a complementary diagnostic of intervention forecasting.

A 20,000-draw model bootstrap gives a Pearson 95% interval of $[-0.15, +1.00]$, excluding degenerate resamples. The pooled ten model–format pairs yield Pearson 0.38 with a model-cluster interval of $[-0.27, +0.78]$. Both intervals include zero, reflecting uncertainty in this five-model comparison.

Figure 5c expresses Capture using the main table's frozen reference normalization. For each setting, completion is $100(s - f)/(c - f)$ with $f = -1$ and that setting's Capture ceiling c. The two settings are then averaged equally, without additional clipping. Models follow increasing main-table SWFI, with connecting lines between categorical labels. Figure 5b uses the same normalization before averaging formats.

Protocol-dependent gains. Beyond the gains in Section 6, weaker models often lose Capture under structured protocols, so process supervision should be adapted to model capability. Comparisons use the same anchors within each model, with the smaller Sol sample and Code's execution and feedback resources described above.

State accuracy and effect accuracy can move differently. Sol's Table exceeds Code on the measured

Table 28: Final-answer scores averaged equally over Omni-ID and OOD. Each model uses 190 anchors, and GPT-5.6 Sol uses 30. Protocols use the same anchors within each model.

Model	Protocol	QUANTITY	EVENT	Capture
Qwen3.8-27B	Free	+0.206	+0.012	−0.408
Qwen3.8-27B	Table	+0.171	+0.009	−0.456
Qwen3.8-27B	Code	+0.127	−0.087	−0.642
Gemma 4 31B	Free	+0.153	+0.009	−0.423
Gemma 4 31B	Table	+0.120	+0.007	−0.443
Gemma 4 31B	Code	+0.015	−0.129	−0.605
Flash-Next	Free	+0.204	+0.015	−0.428
Flash-Next	Table	+0.150	+0.008	−0.427
Flash-Next	Code	+0.061	−0.089	−0.563
Kimi K3	Free	+0.283	+0.012	−0.355
Kimi K3	Table	+0.298	+0.021	−0.354
Kimi K3	Code	+0.213	−0.029	−0.535
GPT-5.6 Sol	Free	+0.335	+0.013	−0.533
GPT-5.6 Sol	Table	+0.345	+0.013	−0.303
GPT-5.6 Sol	Code	+0.256	−0.039	−0.240

Table 29: Associations between process and final-answer scores. Rows use L3 state skill and Capture unless marked L1. Model means average Table and Code before computing correlations. Completion follows the frozen main-table normalization. These are descriptive associations.

Aggregation	Points	Pearson r	Spearman ρ
Model–protocol pairs	10	+0.384	+0.539
Table only	5	+0.556	+0.300
Code only	5	+0.308	+0.800
Model means, raw Capture	5	+0.699	+0.800
Model means, Capture completion	5	+0.693	+0.800
Model means, L1 process and L1 completion	5	+0.872	+0.800

intermediate-state skill, while Code achieves better final Capture. Kimi exhibits the reverse pattern on intervention branch-state skill and Capture. Thus, process supervision should target both states and intervention differences, with downstream Capture evaluated on held-out worlds.

J Negative design results

Report reconstruction and short horizons. Generating an entire future report conflates dynamics with serialization and copying unchanged fields. The benchmark instead scores explicit quantities and events. Very short observational horizons also leave many quantities unchanged, favoring persistence. Longer budget-specific horizons were retained for L1, while L3 retains short horizons when an operator's immediate effect would otherwise fall outside its declared scoring band.

Hidden-target identifiability. Early absolute trait bins were weakly recoverable even from privileged behavioral features, motivating within-world rankings. Several world parameters were removed as inference targets because their effects were absent or ambiguous in the rendered window. Illness illustrates this distinction: it affects engine dynamics but is neither explicitly rendered nor narrated, so city-size changes alone cannot identify its cause. Growth and research questions now require relevant observed changes. These exclusions concern the available evidence, not causal inactivity.

Intervention persistence and observability. Some savegame fields are recomputed by the AI after reload, so a successful file edit does not establish a persistent intervention. Government changes are therefore induced indirectly through technology access. Wonder-grant and composite-trait candidates were removed or replaced when their scored effects were weak or their descriptions did not identify the complete edit. The retained single-trait operators modify one dimension at a time. Attitude interventions remain difficult because changed attitudes need not alter the scored war or alliance counts. This is a limitation of downstream observability, not evidence that attitudes are irrelevant.

Sections E and G specify the resulting rules and references.

K Engine-field semantics and validation

Unit counts. Total unit counts include civilian units. Military and civilian counts are therefore reconstructed separately from unit rosters and ruleset attack values.

Research accounting. Research progress records accumulated points, not one-turn research production. Team games further index research by team rather than player slot. The renderer and truth builder use this mapping and compute technologies learned from known-technology counts, excluding repeated Future Technology advances.

War preparation. The diplomatic countdown is not uniformly a war timer. The renderer accepts values from 1 to 12 for active preparations and verifies the relationship and target before reporting a war plan.

Event time and identity. Engine events and state-difference events use different timestamps. A difference attached to state t describes the transition during turn $t-1$, whereas an engine event at t occurs during turn t. The builder normalizes this convention before constructing histories and targets. It also normalizes recoverable name-encoding errors, excludes destroyed players from relationship counts, and removes illness suffixes from displayed ruleset names.

L Prompt examples

L.1 Shared context and forecast format

Prompts combine observation frames, interval narratives, the agents' current policies where permitted, and a task block. The excerpts below use turn-60 questions from a three-civilization world with a 24×32 map, the civ2civ3 ruleset, and hard AI difficulty. Report rows are omitted for space, and ground-truth metadata are excluded from model inputs.

```
===== OBSERVATION FRAME 1/3 turn 20 (with map) =====
## WORLD REPORT
turn 20, year -3050
ruleset civ2civ3
[map, civilization, city, unit, and relationship rows omitted]
[subsequent narratives and frames at turns 40 and 60 omitted]
```

The quantitative output grammar is unchanged across viewpoints. For this global-view example, three civilizations, eleven quantities, and two horizons require 66 lines.

```
<civilization> | <quantity> | +<horizon> | p10 | p25 | p50 | p75 | p90
Civilizations (3): Hannibal Barca, Leonidas, Hatshepsut
Horizons (2): +60, +120
```

Under the egocentric condition, the report is headed `WORLD REPORT (as Hannibal Barca believes it)` and unexplored map tiles are marked ?. At this anchor the viewer has not contacted the other civilizations, so the question scope contains Hannibal Barca alone. Hidden truth is retained for scoring but is not appended to the prompt.

L.2 Events, hidden targets, and interventions

Binary event items state their empirical reference probability. One retained question asks whether Leonidas forms a new alliance between turns 60 and 180 and supplies a base rate of 0.26. Leader questions instead request a distribution over candidate civilizations. Trait responses use a ranking grammar, such as `Aggressive: <civilization> > <civilization> = <civilization>`. Rule responses use `<setting> | <label>`, with only evidence-eligible settings requested. These strings illustrate syntax, not example correct answers.

The real government-enabling intervention at this anchor states:

> Hannibal Barca acquires the technology The Republic.

The numerical question then requests the same forecast format at the operator's eligible horizon. The corresponding no-intervention prompt supplies the model reference, while the recorded factual rerun supplies the outcome reference. A technology grant creates the opportunity for an AI government change rather than guaranteeing that change.

L.3 Verifiable Table and Code outputs

Table prepends prescribed intermediate predictions to the final answer. Numeric claims use `V | <civilization> | <quantity> | <turn> | <prediction>`. Other supported claims concern diplomatic relations, governments, technologies, city ownership, and wonders. The aligned study specifies required numeric cells before inference, so a model cannot choose only easy claims for its process score.

Code defines `predict(world, rng)` and returns `{"trajectory": {turn: {civilization: state}}}`. The following schematic shows the execution interface, not a fitted model or a supplied answer.

```
def predict(world, rng):
  state = initialize(world["frames"])
  trajectory = {}
  for turn in forecast_turns(world):
    state = transition(state, world, rng)
    trajectory[turn] = snapshot(state)
  return {"trajectory": trajectory}
```

All helper functions and dynamics in a submitted program are generated by the model. The harness executes 50 sampled trajectories, derives final quantiles or event probabilities, and checks intermediate predictions against reserved engine states. Repair feedback uses execution errors and observed-history back-tests, never future evaluation truth. Section I specifies process scoring, and Section O reproduces the mechanism cards.

M Reproducibility

Reproduction uses the fixed prompts and reference values, with 1,900 decision points for each open-weight model, a shared stratified 190-anchor sample for Opus 5, GPT-5.6 Sol, and EXPERT, and 60 shared anchors for additional external-validation models. Run open-weight models through vLLM with native chat templates and thinking enabled, using temperature 1.0, top-p 0.95, and top-k 20. Reserve 16,384 reasoning tokens, or 65,536 for DeepSeek, plus the larger of 1,024 tokens and the task-specific answer estimate. The context window is 262,144 tokens, except gpt-oss at 131,072, with the output allowance reduced when needed to fit. Qwen models use medium effort, Kimi uses high effort, and Inkling uses an NVFP4 checkpoint. Both main API models use medium effort, with adaptive thinking for Opus. Section G.2.4 specifies the expert manual and harness. Nemotron 3 Ultra returned unparseable intervention answers on 33–43% of its required cells, and Inkling on 7–14%. The failure penalties in Section F produce Nemotron's negative Direction and Specificity scores. The benchmark engine image is `freeciv-server:3.1.6-v2`, and the agent-play image adds the task-persistence patch.

Keep reserved engine states and future outcomes outside model inputs. Separate reasoning from final answers, parse each required prediction, and apply the eligibility rules and missing-answer penalties in Section F. Compute component scores and SWFI using the fixed reference scales and weights in Section 4. Retain raw responses and model configurations, and regenerate tables and figures from these outputs without new inference. Sections H.3, I and N specify the process, ablation, and decision experiments, including their sampling and uncertainty procedures.

Table 30: Turn-210 values underlying Figure 6. Differences are paired against the no-WM planning control. $\pm$ denotes standard error across 76 worlds.

World model	Mean score	Δ score	Wins / 76
None	157.49	—	—
Flash-Next	164.00	$+6.51 \pm 5.39$	44
Kimi K3	160.80	$+3.32 \pm 5.93$	39

N Agent-play details

N.1 Scope and protocol separation

Agent play measures decision utility separately from observer-only prediction. The full-game experiment uses 76 selected worlds from turn 60 to 210, with checkpoints at turns 160 and 210. Comparisons share the planning protocol and endpoint, without pooling earlier development variants. The sample includes benchmark worlds and therefore does not test transfer to an independent world distribution.

N.2 Actions and shared information

The actor controls one civilization through structured unit, city, research, budget, and diplomacy commands validated against the engine interface. Observations contain the actor's available information. Refused legal actions, such as treaties, count as outcomes, whereas interface and model-call failures are recorded separately.

All arms share the actor, observations, rules reference, planning opportunities, and execution feedback. The no-WM control retains its own plan. Assisted arms additionally receive forecasts and advice before revision. The adopted plan and discussion remain available during execution, separating WM assistance from ordinary planning support.

N.3 Planning and verification

Planning occurs initially and at intervals of at most ten turns, with earlier replanning allowed after material changes. The agent proposes one to three plans and receives at most three forecasting rounds. Forecasts include score and loss distributions, explicit assumptions, and possible opponent responses. Finalization requires a complete adopted plan. If a final revision differs from the last forecast plan, the logged prediction is not treated as a verified forecast of the revision.

Execution records track positions, city ownership, and garrisons. Resumption retains the saved state and protocol revision, while schema repairs and validation failures are logged. These checks verify execution rather than every natural-language assumption.

N.4 Outcome analysis

Comparisons pair worlds at the same endpoint and report score differences, wins, coverage, and world-level uncertainty. Repeated executions estimate stochastic variation, while forecast calibration and execution logs distinguish prediction from implementation errors. Only completion-validated runs enter the reported analyses.

N.5 Full-game and local outcomes

Table 30 reports positive mean gains at turn 210 across 76 worlds. All 228 runs across the three conditions passed completion validation. Approximate paired t intervals are $[-4.2, 17.3]$ for Flash-Next and $[-8.5, 15.1]$ for Kimi, indicating substantial between-world uncertainty. Runtime repairs and resumption revisions are recorded with the manifest.

Table 31: Local gain relative to self-review, mean ± SE. The 36-world replication re-executes the same finalized plans. The final column pools four executions per plan. The exploratory 12 worlds are excluded from these columns.

WM	First two executions	Next two executions	Four executions
Flash-Next	$+0.39 \pm 1.17$	-0.01 ± 0.78	$+0.19 \pm 0.79$
Kimi K3	-0.15 ± 0.93	-1.17 ± 1.65	-0.66 ± 1.01
Sol	$+1.82 \pm 0.99$	$+0.97 \pm 1.16$	$+1.40 \pm 0.84$
Opus	$+0.96 \pm 1.00$	$+1.61 \pm 1.15$	$+1.28 \pm 0.91$

The local study samples one decision point per world from control trajectories using seed 20260924. The first 12 worlds are exploratory. The next 36 are disjoint from them, with 12 near each of turns 80, 120, and 170. All conditions begin from the same saved state and original proposal within a world. If the historical proposal contains alternatives, the first is retained by a fixed rule. Each self-review or WM condition receives one review/advice opportunity and one agent revision, with the option to retain the original. Revision opportunities are matched, while token budgets may differ. The six finalized plans are executed twice, followed by two additional executions on the 36 validation worlds, totaling 1,008 local branches and 20,160 turns. All passed completion validation. One malformed planning response was retried and retained.

The four WMs forecast the same six finalized plans in two batches of three, with shuffled labels and no author identity or execution results. Forecasts precede the first executions. Flash-Next and Kimi also repeat them before the additional executions. Repeated executions evaluate a fixed plan using stochastic actor calls from the same engine checkpoint without independent engine reseeding. The world remains the independent sampling unit. We use 5,000 world-bootstrap samples with a fixed seed and report individual 95% intervals without multiple-comparison adjustment.

Table 31 reports the 36-world validation sample separately from the 12 exploratory worlds. Four-execution 95% intervals versus self-review are $[-1.38, 1.69]$, $[-2.70, 1.33]$, $[-0.26, 3.00]$, and $[-0.47, 3.11]$ for Flash-Next, Kimi, Sol, and Opus, respectively.

Outcome-informed selection. For each world, we select the plan with the highest mean score in executions 1–2 and evaluate it on executions 3–4, then reverse the roles and average. Tied plans receive their mean held-out score. Expanding the pool from original and self-reviewed plans to all six gains 2.36 ± 0.90, with interval $[0.80, 4.19]$. This post-hoc pool comparison shows that WM-generated plans supply useful alternatives for execution-based selection. The six-plan selector also gains 2.90 ± 0.86 over the original plan and 2.44 ± 0.63, interval $[1.29, 3.78]$, over uniform selection from the same pool. Selection uses actual rollout outcomes, and the expanded pool requires additional generation compute. Forecast-only selection yields gain intervals containing zero for all four WMs. Thus, useful plan generation and reliable forecast-based selection remain distinct capabilities.

Capability comparisons. The paired Opus-minus-Kimi contrast on four executions is 1.94 ± 0.76 with interval $[0.55, 3.54]$, and Sol-minus-Kimi is 2.06 ± 1.15 with interval $[-0.11, 4.33]$. This exploratory four-model comparison relates OMNI-ID diagnostics to fog-of-war utility. Estimating independent level contributions or fitting index weights requires a larger model panel.

Table 4 reports level completion on OMNI-ID. Each metric is normalized by its frozen floor and ceiling, then averaged equally within its level and multiplied by 100, following Equation (5). L1 combines quantity and event, L2 combines trait and rules, and L3 combines capture, direction, and specificity. Completion percentages are reported without clipping.

Table 32: Component diagnostics and utility accompanying Table 4. Diagnostics are OMNI-ID scores from Table 3, while SWFI spans three settings. Utility averages four executions per plan in 36 worlds and is reported relative to self-review, with world-level SE.

Model	L1 quantity	L2 trait	L3 capture	SWFI	Δ score
Flash-Next	0.26	0.10	-0.57	16.93	$+0.19 \pm 0.79$
Kimi K3	0.34	0.00	-0.35	21.94	-0.66 ± 1.01
Sol	0.27	0.23	-0.60	27.68	$+1.40 \pm 0.84$
Opus	0.33	0.17	-0.08	29.47	$+1.28 \pm 0.91$

N.6 Case study: from accepting collapse to targeted defense

The turn-173 case is selected retrospectively for strategic disagreement and improvement in both execution batches. Tupi has nine cities, score 161, and treasury 83, under threat from Barbarossa's Frigates and Riflemen. Figure 13 traces the original plan, one Opus consultation, one agent revision, and recorded engine feedback.

A decision reversal: accept city losses or attempt targeted defense?

Checkpoint. Turn 173: score 161, 9 cities, 83 gold. One consultation with Opus, followed by a Flash-Next revision and 20 turns of execution.

Agent (Flash-Next): original plan.

"Icaraí and Siri Jibe have empty garrisons and P=1.00 regardless of actions; accept their loss."

"Do NOT build anything at Icaraí or Siri Jibe (likely lost)." "Do NOT buy anything."

World model (Opus): forecast and advice.

Original-plan forecast at T193: score $[p_{10}, p_{50}, p_{90}] = [132, 155, 174]$; $P(\text{any city lost}) = 0.82$.

"'P=1.00' is P(loss GIVEN assault), not P(assault)" [...] "Realistic near-term capture risk is therefore moderate, not certain."

"an empty city is captured by any single unit, a walled city with even one defender forces a fight."

"Reconsider the Capuan triage: it is the most likely city to be assaulted NEXT turn (adjacent Riflemen), and one more fortified Musketeer takes P(loss | assault) from 0.81 to 0.03 per the calculator [...] consider buying it if gold allows (~60-90 gold) rather than hoarding 83 gold that will drip away anyway."

Agent (Flash-Next): revised plan.

"T173 BUY: Buy Capuan Musketeers (20/40 shields remaining)."

"T173 DEFENSE: Order Musketeer 520 from Paráyba (23,2) to Icaraí (21,1)."

Siri Jibe: "switch from Colosseum [45/70 stored] to Musketeers [40]."

Execution feedback (first execution, T173).

```
OK bought Musketeers in Capuan for 60 gold (completes at turn change)
OK Siri Jibe now builds Musketeers (half the stored shields lost: category change)
moved unit 520 Musketeers to 21,1
```

Measured score trajectory (mean of four executions per frozen plan).

Plan	T173	T178	T183	T188	T193
Original	161.00	147.75	127.00	113.50	102.25
Self-review	161.00	149.00	125.00	113.00	109.00
Opus-assisted	161.00	147.25	137.25	123.50	124.25

At T193, the revised plan reduces mean score loss by **22.00 points** relative to the original.

Figure 13: Recorded agent–WM exchange and outcomes in the targeted-defense case. Quotations preserve model outputs, with [...] marking omissions. The forecast concerns the original plan, and 0.81/0.03 are estimates from the shared siege calculator. Recorded executions provide outcome evidence.

Table 33: Complete six-condition results for Figure 13 after 20 turns. Each pair independently executes the same finalized plan. The last column averages remaining city counts across all four executions.

Plan	First pair	Second pair	Mean score	Mean cities
Original	106, 83	109, 111	102.25	1.75
Self-review	86, 147	84, 119	109.00	1.75
Flash-assisted	95, 95	117, 108	103.75	2.25
Kimi-assisted	96, 116	117, 106	108.75	1.50
Sol-assisted	118, 115	127, 139	124.75	3.50
Opus-assisted	134, 114	115, 134	124.25	3.75

Decision and outcome. Table 33 shows a 22-point mean gain and two additional surviving cities for the Opus-assisted plan. Icaraí survives three of four executions, versus none under the original or self-reviewed plan. The agent translates advice into a Musketeer relocation. The plan still loses most cities, and the original-plan forecast $[132, 155, 174]$ exceeds the realized 83–111. The benefit comes from the revised plan as a whole and illustrates useful strategic advice despite forecast error.

N.7 Failure case: precaution without realized benefit

The turn-172 case supplies a contrasting example under the same local protocol. The original plan emphasizes research. Sol forecasts only a 0.07 probability of losing any city within 20 turns, but recommends earlier tax increases and additional defenses. The agent adopts those precautions and raises the treasury trigger from the adviser's 300–350 gold to roughly 400–450 in its revised plan. Figure 14 shows the recorded exchange and executed changes.

At T192, the four original executions score 175, 194, 175, and 174, compared with 173, 171, 170, and 172 under Sol assistance. The mean difference is -8.00 points, with -12.5 in the first batch and -3.5 in the replication. The self-review mean is 177.75, also above the assisted mean of 171.50. All three conditions retain eight cities throughout. Relative to the original, the assisted plan produces fewer research bulbs per turn, 46.3 versus 56.4, finishes with fewer technologies, 35.25 versus 36.50, and retains more gold, 423.5 versus 386.0. The original executions never end a turn below 358 gold.

The agent follows and strengthens the precautions, incurring a resource cost without a realized survival benefit. The score gap reflects the combined plan revision, including taxation and defense changes. Together, the retrospectively selected cases motivate state-specific advice that balances strategic risks against opportunity costs.

Evidence verification. Quoted advice and plans are checked against recorded consultations, and outcomes against engine trajectories and execution feedback. The main gain plot clips confidence bands to $[-1.5, 7.5]$ for readability, while the data retain their full bounds. These are pointwise, not simultaneous, intervals.

A failed revision: precaution without a realized score benefit

Checkpoint. Turn 172: score 148, 8 cities, 665 gold. One consultation with Sol, followed by a Flash-Next revision and 20 turns of execution.

Agent (Flash-Next): original plan.

"PRIMARY LEVER: tech at 69 bulbs/turn."

"Treasury 665, -13/turn, ends ~T210 at ~160"

World model (Sol): forecast and advice.

Original-plan forecast at T192: score $[p_{10}, p_{50}, p_{90}] = [158, 168, 178]$; $P(\text{any city lost}) = 0.07$.

"Begin raising taxes if the treasury falls below roughly 300-350 or if the observed deficit remains above 10 gold/turn; waiting for 200 risks forced building sales."

"Oviedo and Santander particularly benefit from City Walls or a second defender"

Agent (Flash-Next): revised plan.

"Must raise taxes to 60/30/10 by ~T182-185 when treasury nears 400 to prevent forced building sales."

"DEFENSE: City Walls (0 upkeep, +100% defense) on Oviedo, Santander, Valencia"

Execution feedback (first execution).

T176: `OK rates tax 60% science 30% luxury 10%`

T177: `OK Valencia now builds City Walls`

T179: `OK Oviedo now builds City Walls`

Measured score trajectory (mean of four executions per frozen plan).

Plan	T172	T177	T182	T187	T192
Original	148.00	153.50	159.50	172.00	179.50
Self-review	148.00	153.00	162.75	170.00	177.75
Sol-assisted	148.00	152.50	156.25	164.75	171.50

The revision yields **8.00 fewer score points** at T192. All three conditions retain all 8 cities throughout. Mean science output is 56.4 versus 46.3 bulbs/turn (original versus Sol-assisted); final treasury is 386.0 versus 423.5. The original executions never end a turn below 358 gold.

Figure 14: The precautionary revision retained more gold but reduced research output and final score. Quotations reproduce logged advice and plans. The forecast concerns the original plan, while outcomes reflect the complete revision.

O Mechanism cards and the direct-effect check

Mechanism-card protocol. The eleven cards derive from engine source, patches, and the expert manual. Checks exclude civilization names from the evaluated anchor and verify consistency with the engine. Each operator uses the same text across worlds and real/placebo arms, with edit magnitudes and outcomes withheld. The following paragraphs reproduce the intervention-specific prompt text verbatim. Section H.4 specifies the paired comparison and coverage.

Independent direct-effect check. The diagnostic examines the target civilization's directly affected quantity at the shortest asked horizon for four visible-channel operators. It compares the sign of the model median's shift from its own control with the realized arm-minus-factual sign, and separately records zero responses and the absolute effect-magnitude ratio. Flash-Next's sign agreement is 0.77 / 0.75 / 0.78 across the three settings, with zero responses on 2–4% of cells and median magnitude ratios of 1.1–1.5. These are conditional summaries, not Capture scores. The diagnostic includes city count outside the main scoring mask, and median ratios respond differently to outliers than Capture. It therefore complements the channel analysis in Section H.4 without replacing it.

Treasury (gold stock). The treasury is the civilization's stock of gold. Every turn the engine adds income (trade times the tax rate, plus trade-route revenue and the Palace bonus) and subtracts gold upkeep for buildings and, under Despotism, Monarchy, Fundamentalism, Democracy and Federation, for units beyond the free ones per city

(Tribal, Republic and Communism pay unit upkeep in shields instead). When the treasury cannot cover upkeep the engine sells a building or disbands units, and the AI reacts within a few turns by moving rate points from science to tax and by stopping rush-buys; a large treasury lets it keep science high, rush-buy defenders in threatened cities and pay for bribes and incitements. The edit changes the stock once; income and upkeep rates are unchanged, so the difference is worked off only through the AI's own responses (rate points moved between tax and science, rush-buying or not), typically over some tens of turns, during which the treasury drifts back toward its trend while the number of units and buildings moves slightly in the same direction as the edit. It does not change technologies known, city sizes, terrain, diplomatic relations or personality.

Research points (bulbs). Research points accumulate toward the current research target; when the stock reaches the target's cost the technology is learned, the surplus carries over, and the AI picks the next step toward its research goal. If the added points reach the current target's remaining cost the target is learned at once (and possibly the following ones); otherwise the target simply arrives earlier. Either way technologies arrive earlier by roughly (points added) divided by (points per turn) turns, and the number learned in the window rises by roughly (points added) divided by the current cost, which grows linearly along the tree in civ2civ3 and more steeply in classic. New technologies unlock units, buildings and governments that the AI starts using, add to score, can trigger a government change when a government technology arrives, may be passed to allies by treaty and stolen by neighbours' diplomats. The edit does not change gold, cities, terrain or relations directly.

Destruction of cities. Destroying a city removes it with its citizens and buildings and any wonder inside; units homed there are disbanded or rehomed, its worked tiles and trade routes are freed, and the owner's city count, total city size, population (5 x size x (size+1) per city), production, trade and science drop at once by the destroyed cities' share, disproportionately so when the destroyed cities are large ones, since population is convex in size; culture loses the destroyed cities' own accumulated culture while the national history is kept. The AI does not rebuild quickly: Settlers are only built where the founder want is high, so the count recovers at the civilization's normal expansion pace if free land is within reach and otherwise not at all, and neighbours may settle the freed land. Score falls with the lost citizens and later with the slower technology pace. Its standing as a war target moves in both directions: fewer and smaller cities lower the value the AI's target formula assigns to it, while fewer defending units lower the fear term, so neighbours may or may not become more inclined to attack. Personality, technologies known and diplomatic states are not changed by the edit.

A technology is granted. Granting a technology adds it to the civilization's known set without spending research points; if it was the current research target the accumulated points carry to the next target, and if it was a prerequisite the path to the research goal shortens. If the technology enables a government, the AI re-evaluates every available government every few turns (about 5 to 20, growing with the city count, and immediately after a reload) and switches when the new one scores higher for its situation, so the change follows the enabling technology within that lag and passes through one to five turns of Anarchy (no taxes or science, culture stalls) before the new rules take effect: for example Communism pays no gold for 1-coin buildings and has no distance-based corruption but a lower rate cap (80%) than Democracy (90%); Republic adds trade and pays unit upkeep in shields, while Democracy adds the same trade but pays doubled gold upkeep for units, and both suffer war weariness. A technology that unlocks units or buildings changes what cities build over the following 10 to 30 turns, and allies may receive it by treaty. The edit does not change gold, cities, terrain or relations directly.

Map knowledge revealed. Revealing map tiles gives one civilization knowledge of the revealed tiles' terrain and of the cities standing on them at that moment; what it later sees is again limited to its vision. The AI uses map knowledge for settling (the founder want only considers known free sites), for exploration (explorers stop when nothing is unknown), for routing units and ferries, and for target evaluation (only known cities can be targeted). The effects are indirect and modest: fewer explorer units, possibly earlier or farther settling if free land exists, and earlier contact with distant civilizations, which starts relations at Peace and makes embassies, treaties and later wars with them possible. The edit does not change any quantity of the civilization directly, nor its technologies, relations or personality.

Personality: Aggressive. Aggressive (10 to 150, default 50) scales the AI's appetite for war. In this engine it scales the grudge the AI builds each turn against its preferred war target before declaring (about 5 attitude points per turn at 10, 25 at 50, 75 at 150), raises the attitude threshold below which it declares war so that less hatred is needed (about -360 at 10, -210 at 50, -15 at 150), lowers how much hatred still allows a ceasefire (love above -850 at 10, above -300 at 150), and scales the minimum time after a declaration during which it refuses any ceasefire (about 2 to 4 turns at 10, 10 to 20 at 50, 30 to 60 at 150). An upward shift makes a declaration against the most disliked contact more likely within 20 to 40 turns, in proportion to its size, raises the military share of production and the military unit count, drains the treasury under gold-upkeep governments and worsens the target's relations; a downward shift makes the civilization avoid starting wars, though allies can still drag it in. It has no direct effect on research, growth, production formulas or world rules.

Personality: Expansionist. Expansionist (10 to 150, default 50) scales the founder want with which cities decide to build Settlers and the value the AI puts on free city sites. A higher value means more Settlers and more cities founded per 100 turns, but only where free land within reach exists, since land is the binding constraint; more small cities raise the city count and, slowly, total size, population and score, while spreading units and upkeep thinner. A lower value slows or stops founding while the existing cities grow as before. It does not affect war behaviour,

research, tax rates or relations directly, although a civilization with many Settlers is a slightly more attractive war target in the AI's formula and borders growing into a neighbour's land raise tension.

Personality: Trader. Trader (10 to 150, default 50) scales, linearly in this engine, the AI's want for Caravans built to establish trade routes; caravans for helping wonders are requested by the wonder city's need, although any caravan may be redirected to help a wonder. Trade routes add permanent trade to both cities (routes between a civilization's own cities are allowed here). A higher value shows as more Caravans in the unit list (civilian count up), more trade routes and hence more gold and science over 30 to 100 turns, and possibly faster wonder completion; a lower value the opposite. It does not affect military production, war behaviour, growth or research rules directly.

Personality: Builder. Builder (10 to 150, default 50) discounts the military advisor's want where it competes with the domestic advisor in each city (about x1.67 at 10, x1.0 at 50, x0.5 at 150), except when a city is in danger or the civilization is on a war footing, and scales the building wants themselves in the same direction (about x0.5 at 10, x2.3 at 150). A higher value shifts production from units to buildings and wonders: fewer military units per city, more improvements (and with them culture, science and gold multipliers) and somewhat faster score growth; a lower value the opposite. It does not directly change war declarations, research, growth or relations, though a smaller army makes the civilization a somewhat more attractive war target.

Hidden attitude (love) toward one player. Love (-1000 to +1000) is the AI's hidden attitude toward one other player, recomputed every turn: about +15 per turn in any non-war state (+10 more when allied, +7.5 per shared enemy, and +60 more while a ceasefire holds), -15 per turn at war plus penalties for that player's units inside its borders, a 4 percent decay toward zero, and the grudge it builds toward its chosen war target. It gates diplomacy: war is declared when love falls below the aggression-dependent threshold and the player has been chosen as target; a ceasefire needs love above the ceasefire threshold, a peace treaty above +125, an alliance above +250 from Peace with no war countdown. The edit shifts the stock once and the per-turn dynamics then pull it toward the level the current relation supports, so a large drop can start a war countdown or a declaration against that player within a few to a few dozen turns (and the target's relations and enemy count change accordingly), while a large rise can lead to a ceasefire or peace if at war, or to an alliance from Peace if love stays high. It does not change the other player's attitude, nor any economic quantity directly.

World rule: disaster frequency. The disaster setting is a world rule: the per-city, per-turn chance of a random disaster (Earthquake or Fire destroys a building, Flood costs food or a citizen, Plague costs a citizen, Famine empties the granary, Industrial Accident creates pollution). Multiplying the setting multiplies the expected number of such events proportionally in every city of every civilization from the edit onward; each event costs a building, a citizen, a granary or a polluted tile, so total city size, population and culture grow slightly more slowly and production is diverted to rebuilding. The effect is world-wide, small per event and accumulates with city count and time; it does not change relations, research rules, personality or tax rates.